IHP SG13G2 · OPEN SOURCE RFIC EM WORKFLOW

EM Simulation from KLayout: User's Guide for the klayout_em_export Plug-in

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Document version 2026-11-04, plug-in version 0.2.1

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1 What the plug-in does

In this chapter: what the plug-in is for, and how it fits between your layout and your circuit simulator.

Why it matters: at mm-wave frequencies the wiring around a transistor is part of the circuit. Its inductance, capacitance and coupling change gain, matching and stability. The plug-in lets you put exactly that wiring into an EM simulation, and then put the transistors back in a circuit simulator.

An EM simulator such as Palace computes the electromagnetic behaviour of metal and dielectric. It cannot simulate a transistor. A circuit simulator such as ngspice simulates the transistor with its PDK model, but knows nothing about the wiring you drew. For a mm-wave amplifier you need both.

The plug-in connects the two worlds:

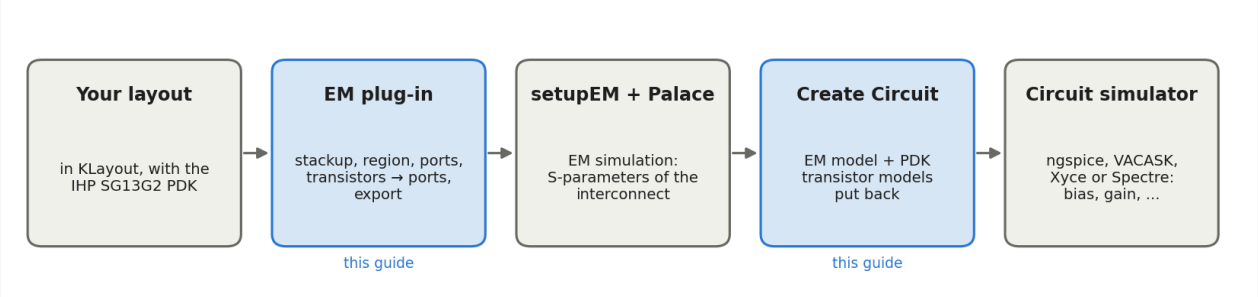


Figure 1: The workflow

1. **In KLayout** you open your layout and mark what to simulate. The plug-in replaces each transistor by ports at its pins, places ports at the pads, and exports a clean layout for the EM simulation.
2. **Palace** (set up in setupEM or EMStudio) or **openEMS** (set up in EMStudio) simulates that layout. The result is an S-parameter file: the interconnect, seen from all ports.
3. **Create Circuit** builds a circuit from it: the EM result, with the PDK transistor models connected again at their ports.
4. **Your circuit simulator** (ngspice, VACASK, Xyce or Spectre) then simulates bias, gain and everything else, with the real wiring included.

This guide follows the Palace flow with setupEM, the one that puts the transistors back. For openEMS, see EMStudio instead of setupEM in chapter 11. The same layout can also go to a thermal simulation with setupThermal, for the temperature of each device: chapter 14.

1.1 What the plug-in does for you

Much of the work that makes an EM model tedious, and easy to get wrong, is done for you:

- **A via port in one click.** Click a metal edge: the port is placed exactly along that edge, and the port dialog offers only the metals that are really drawn there (chapter 6).
- **An in-plane port in one click.** Shift+click into a gap: the plug-in finds the gap between two shapes of one metal, draws the port across it and sets its direction and metal (chapter 6).
- **Ground found and checked.** The plug-in traces your metal through the vias, finds the ground net by itself and shows it in the layout. It warns you before the simulation if a port is shorted, does not reach ground, or if your layout has no ground at all (chapter 7).
- **Ports where the region cuts your wiring.** Simulate only part of the layout, and the plug-in proposes a port on every signal wire that leaves the region, from the wire to the ground metal there. Ground wires get no port (chapter 8).
- **Transistors replaced by ports.** The plug-in recognises the SG13G2 transistors and resistors, finds their terminals from the PDK cell (for a MOSFET it follows the gate to your contacts) and places one port per terminal, over all fingers. The reference patch covers only the device's pins, so two devices side by side keep

separate references. The export removes the device's own metal below the pins, so nothing is counted twice: once in the EM model and once in the PDK model (chapter 9).

- **Connection checks for devices.** You are warned if a device terminal is not wired to anything in the region, or if the reference patches of two devices touch (chapter 9).
- **A clean export.** One flat cell, cut at the region, with only the layers the stackup needs, and a report of everything left out. Ports are numbered 1, 2, 3, … without gaps, and all via ports point the same way, with the signal metal at the same end (chapter 10).
- **Back to the circuit in one step.** The export records which port belongs to which pin of which device. Create Circuit uses it to connect the PDK models again and writes the circuit for your simulator (chapter 12).
- **Settings kept with the layout.** The stackup, the port settings and your ground marks are saved in the GDS file. The EM panel shows with green, orange and red marks what is done and what is still missing (chapter 4).

This guide follows one example from start to end: a cascode core with two npn13G2 transistors, a bias divider of three resistors, and two RF pads.

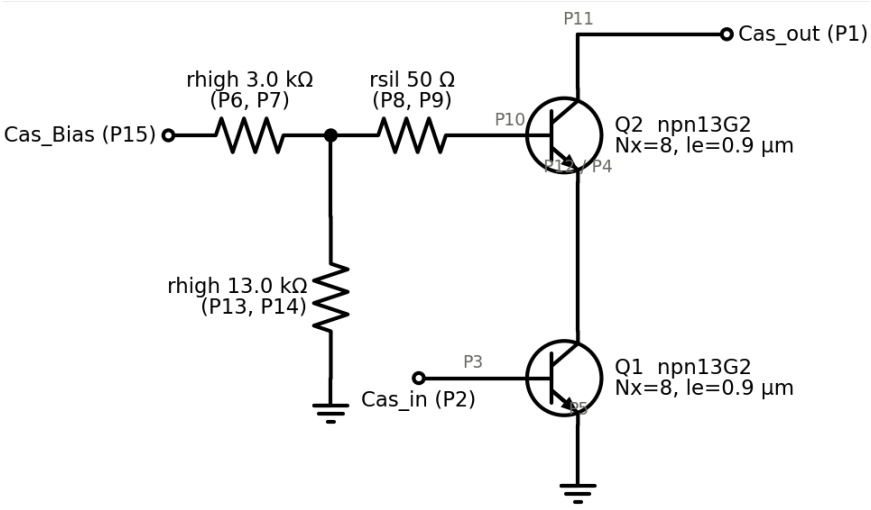


Figure 2: The example: a cascode core

In the layout, the two transistors sit stacked in the middle, Q2 above Q1. The RF lines Cas_in and Cas_out leave at the bottom and top, the bias resistors are at the left edge, and a large ground plane surrounds everything. This is the layout you will prepare for the EM simulation in the following chapters.

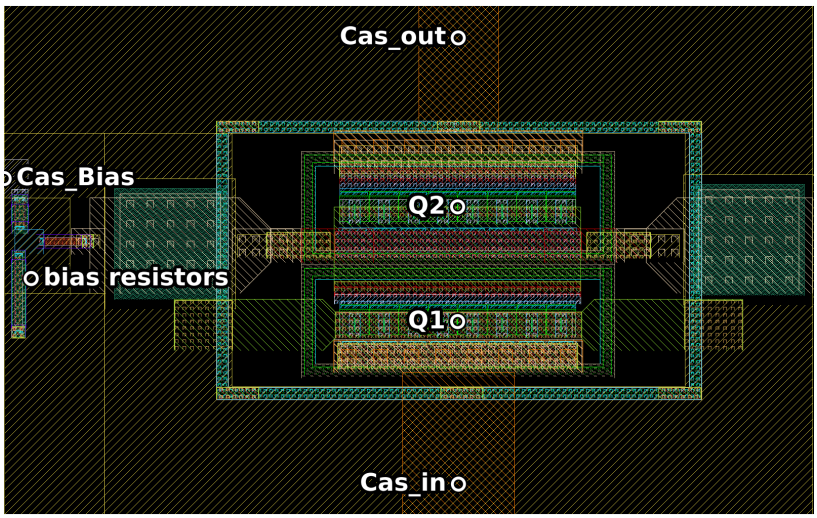


Figure 3: The cascode layout (SG13G2), about 51 µm × 32 µm

2 Installation

In this chapter: the software you need, and how to check that everything is in place.

Why it matters: the plug-in works together with other tools: the IHP PDK in KLayout recognizes your transistors, gds2palace and setupEM build the EM model, and snp2le turns the result into a circuit.

You need:

What	What for	How to get it
KLayout 0.29 or newer	the layout editor the plug-in runs in	klayout.de
IHP SG13G2 PDK for KLayout	recognizes your transistors and resistors (PDK cells)	IHP-Open-PDK, KLayout setup as described there
the plug-in	this guide	copy the `klayout_em_export` folder into KLayout's package folder (below)
a Python environment with setupEM	setupEM, gds2palace, snp2le	`pip install setupEM snp2le` (setupEM installs gds2palace)
Palace	the EM simulator, on Linux	see the gds2palace documentation
a circuit simulator	ngspice, VACASK, Xyce or Spectre	ngspice, VACASK and Xyce come with the IHP PDK models, e.g. in the IIC-OSIC-TOOLS image

Installing the plug-in:

1. Close KLayout.
2. Copy the `klayout_em_export` folder into KLayout's package folder:
   - Windows: `%USERPROFILE%\KLayout\salt\`
   - Linux: `~/.klayout/salt/`
3. Start KLayout.

Check: the menu bar now has an **EM** menu, the toolbar an **EM Port** button, and the **EM Export** panel opens on the right.

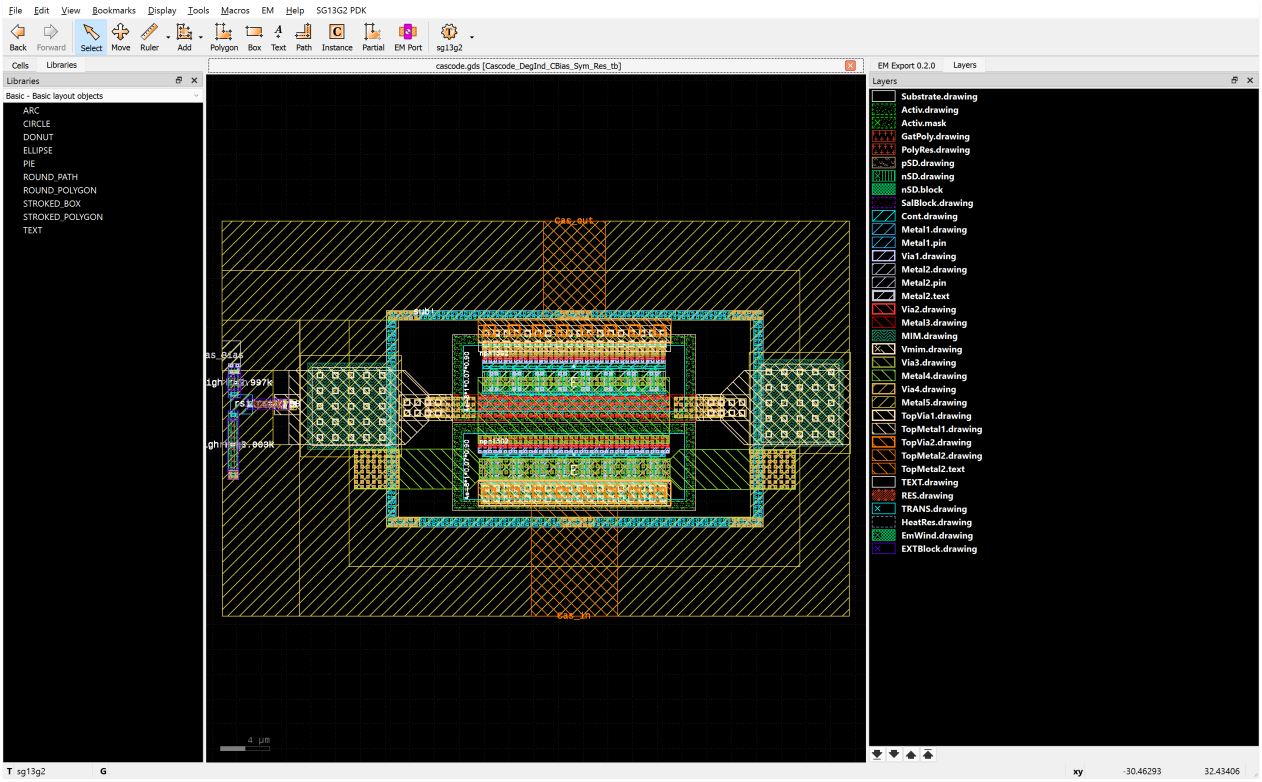


Figure 4: KLayout with the plug-in: EM menu, EM Port button and the EM panel

What can go wrong: no EM menu means KLayout did not find the plug-in: check that the folder is directly in `salt` and contains `grain.xml`. If transistors are not recognized later (chapter 9), the IHP PDK is not loaded in KLayout.

3 A first run in 15 minutes

In this chapter: the whole flow in short, without explanations, so you see it work once.

Why it matters: with the big picture in mind, the following chapters are easier to place. Each step here points to the chapter that explains it.

1. Open your layout in KLayout. Choose the stackup: **EM > Choose Stackup…**, e.g. `SG13G2_FEM_200um.xml` from gds2palace (chapter 5).
2. Place a port at each pad with the **EM Port** tool: click the edge of the pad metal (chapter 6).
3. Check the **Ground** line in the EM panel: it should show the ground metals with a green check (chapter 7).
4. In the panel's **Devices** line, open the details and press **Replace All** (chapter 9).
5. Press **Export for EM…** and **Export** (chapter 10). setupEM opens with your model.
6. In setupEM, set the frequencies, starting at **0 GHz**, create the model and run Palace (chapter 11).
7. Back in KLayout: **EM > Create Circuit (EM Result with Devices)…**, choose the result `_dc.sNp` and your simulator (chapter 12).
8. Include the circuit file in your testbench and simulate (chapter 12).

4 The EM menu, the EM panel and the EM Port tool

In this chapter: where to find the plug-in's functions.

Why it matters: most of the work happens in the EM panel. It works like a checklist: a green check on each line means that step is ready.

The EM menu has all functions, top to bottom in the order you use them:

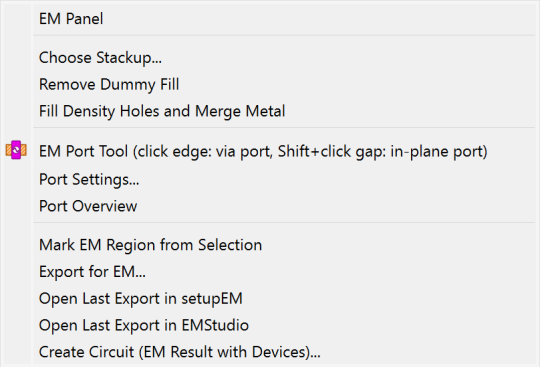


Figure 5: The EM menu

The EM panel (EM > EM Panel, if you closed it) shows the state of your layout:

- **Stackup**: the stackup file in use.
- **Region**: what will be exported: the whole cell, your selection, or the EM region box.
- **Ground**: the ground net and the checks on it.
- **Devices**: the transistors and resistors found, and how many are replaced by ports.
- **Ports**: all ports, as a table.

Each line has an arrow on the right. It opens the details, with the buttons for that step. Below the table are the **EM Port tool**, **Create Proposed Ports** (chapter 8) and **Refresh**, and at the bottom **Export for EM…** and **Open in setupEM**. The switch at the top, **EM | Thermal**, changes the panel to the thermal export (chapter 14).

The marks mean:

Mark	Meaning
✓ (green)	ready
⚠ (orange)	works, but check the details
✗ (red)	missing: the export will not work, or not as intended

The EM Port tool is the **EM Port** button in the toolbar. With it, a click on a metal edge creates a port (chapter 6).

5 The stackup

In this chapter: choosing the stackup file.

Why it matters: the stackup tells the EM simulator the height, thickness and material of every metal and dielectric layer. Every later step uses it: which metals can carry a port, how vias connect metals, which layers are exported.

At the start, the panel shows a red mark at **Stackup**:

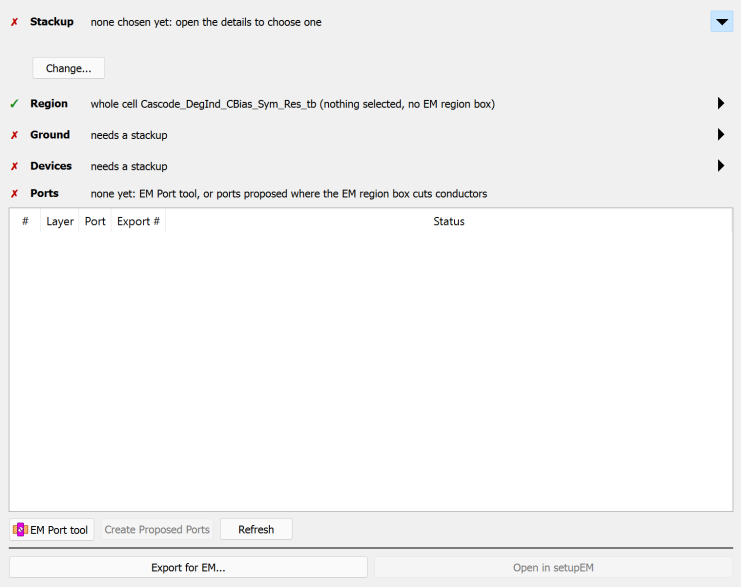


Figure 6: No stackup chosen yet

1. Open the details of the **Stackup** line (arrow on the right), or use **EM > Choose Stackup...**.
2. Press **Change...** and choose the stackup XML, e.g. `SG13G2_FEM_200um.xml` from gds2palace (`XML_stackup/latest/IHP/`).

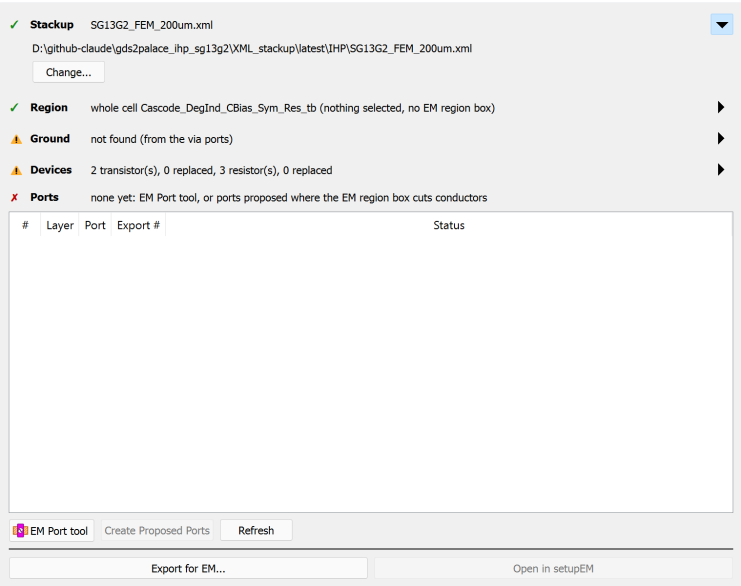


Figure 7: Stackup chosen

The stackup is stored with your layout, and used as the default for the next layout you open.

What can go wrong: use a stackup made for your simulator: from gds2palace for Palace, from gds2openEMS for openEMS. They look similar but contain different modelling assumptions.

5.1 Dummy metal fill

A layout ready for tape-out is full of dummy metal fill: tens of thousands of small tiles that clutter the view and are hard to hide. **EM > Remove Dummy Fill** removes them from the current cell, in one undo step (**Edit > Undo**).

- **What counts as fill:** a shape on a metal of the stackup, on any datatype, that floats and repeats. It floats when nothing connects to it: no via of the stackup, no label, no port, and it is not part of a device. Markers on other datatypes, such as a no-fill area, do not count: they are no metal. It repeats when its size is found at least 20 times among the floating shapes. This works for any fill tool, whether the fill is drawn on its own datatype or on the drawing datatype, flat or as a cell placed many times.
- **Drawn metal** (datatype 0) is part of your EM model, so it is treated more carefully: a drawn shape that touches a shape of any other datatype, a marker for example, stays.
- **What stays:** all vias and via arrays (via layers are not searched at all), every shape that a via, a label or a port touches, a single floating shape (a shield, a test structure), devices, and the layers that are not metals of the stackup (e.g. GatPoly fill stays).
- A fill tile cell loses the placements that are fill. If none is left, the cell is deleted.
- Fill on the drawing datatype is floating metal in the EM model: removing it removes its coupling capacitance. Fill on its own datatype is not exported anyway; removing it only cleans up the view.
- Large layouts take a while: about 60 s for 600,000 fill shapes.

5.2 Cheesed metal and many small shapes

Density rules put many small openings into wide metal (“cheesing”), and a layout is often drawn from many overlapping pieces: via stacks with their own metal pads, tiles, lines. To the EM fields this is solid metal, but every opening and every piece adds edges that the mesh must follow, and the plug-in’s checks get slower. Convert the layout once, before you place ports: **EM > Fill Density Holes and Merge Metal**.

- **Holes:** on every metal of the stackup (not the vias), an opening size found at least 20 times is cheesing and is filled, in any shape. Openings of other sizes belong to your circuit and stay.
- **Merge:** then the metal of the current cell is merged into as few polygons as possible. The metal of subcells placed at least twice (via stacks, MIM capacitors) is merged in too, and so is the metal of a subcell that holds filled holes (a cheesed plane in a cell of its own).
- **What stays:** devices, PDK cells, other subcells, cells that are also placed somewhere else, all vias (the export writes every via, and gds2palace corrects the via arrays itself), texts and the plug-in’s own marked shapes.
- A message lists per metal the holes filled and the polygons before and after. On the 60 GHz PA of the examples: 2899 metal polygons before, 387 after.
- **Edit > Undo** takes it back. Once you save and close the layout, the change stays: **never tape out a converted layout**, the fill breaks the density rules. Convert a copy (File > Save As first) if you are unsure.

6 Ports

In this chapter: what an EM port is, and how to place one.

Why it matters: the S-parameters are only as good as the ports. A port defines where the EM simulation feeds signal into the layout and measures it, and between which two conductors. A port in the wrong place or with the wrong reference gives you a wrong reference plane, or a short circuit.

6.1 What is a port?

A port is a small source between two conductors: the signal conductor and its reference (ground). The plug-in uses two kinds:

- **Via port:** a vertical port between two metal layers, e.g. from the pad on TopMetal2 down to the ground plane on Metal5. You draw it as a line (zero width) along a metal edge. This is the usual port at a pad.
- **In-plane port:** a horizontal port across a gap between two shapes of the same metal, e.g. in a coplanar line. You draw it as a box that bridges the gap.

Each port is a shape on its own layer: port 1 on layer 201, port 2 on layer 202, and so on. Its settings (direction, layers, impedance) are stored on the shape, so they are saved with your layout.

6.2 Placing a port with the EM Port tool

1. Press **EM Port** in the toolbar.
2. Move the mouse over the edge of the metal where the port should be. The edge under the cursor is highlighted.
3. Click. The port shape is placed, and the port dialog opens.
4. Check the settings and press **OK** (**Cancel** removes the port again).

For an in-plane port, hold **Shift** and click into the gap between two shapes of one metal.

The tool picks only metals you can see. Hide a layer in the layer panel, and its edges are ignored, so you can reach a metal edge below a larger shape. If the only edge under the cursor is on a hidden layer, the status bar tells you. Hiding a layer does not limit the port itself: the port dialog still offers every metal drawn there, e.g. the ground plane you hid to see the pad.

In the cascode, the two RF pads get a via port each, at the outer edge of the pad (port 1 at the Cas_out pad):

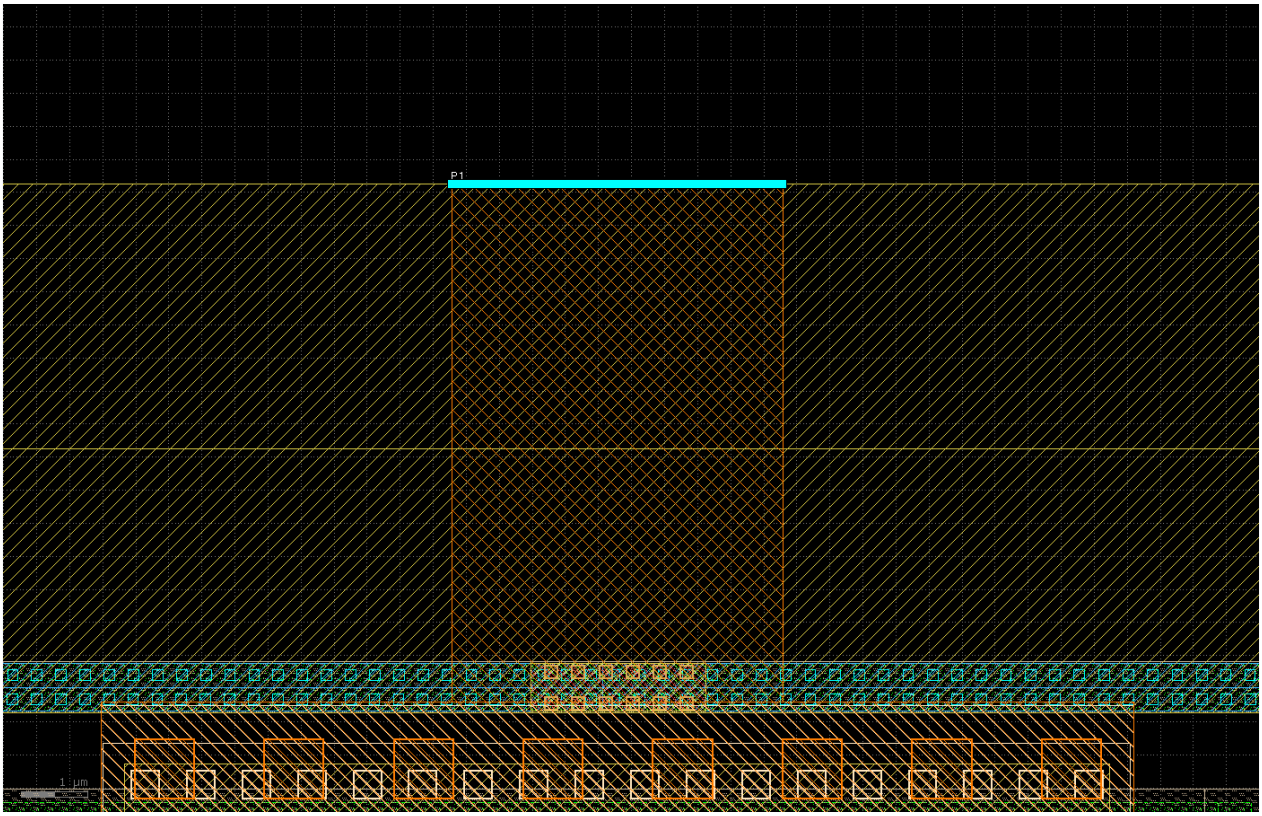


Figure 8: Port 1 at the edge of the Cas_out pad (cyan line)

6.3 The port dialog

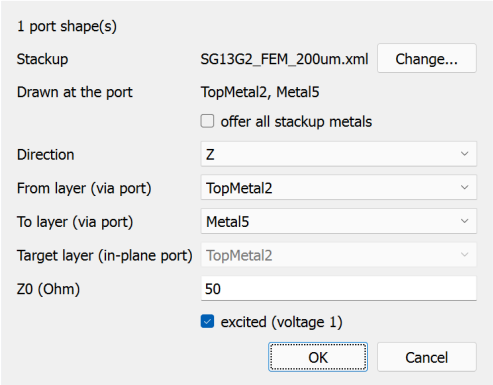


Figure 9: The port dialog

- **Drawn at the port**: the metals found at the port's location. Only these are offered, unless you tick **offer all stackup metals**.
- **Direction**: `Z` for a via port, `X` or `Y` for an in-plane port. You do not need `-Z`: the export chooses the sign of each via port itself (chapter 10).
- **From layer / To layer** (via port): the signal metal and the reference metal. For the pad: from TopMetal2 (pad) to Metal5 (ground plane). A new via port starts from the metal you clicked, down to the next metal below it at the port: a via port stops at the first metal it touches. If nothing is below, it goes to the next metal above.
- **Target layer** (in-plane port): the metal with the gap.
- **Z0**: the port impedance, usually 50 Ohm.
- **excited**: whether the simulation feeds signal into this port. Leave it ticked.

To change a port later: select its shape and use **EM > Port Settings...**, or double-click its line in the panel's port table. A port shape you drew yourself, without settings yet, starts like a new port: from the top metal at the port down to the next metal below. If only one metal is at the port, the dialog offers all stackup metals.

6.4 Checking your ports

The panel's table lists all ports with their status:

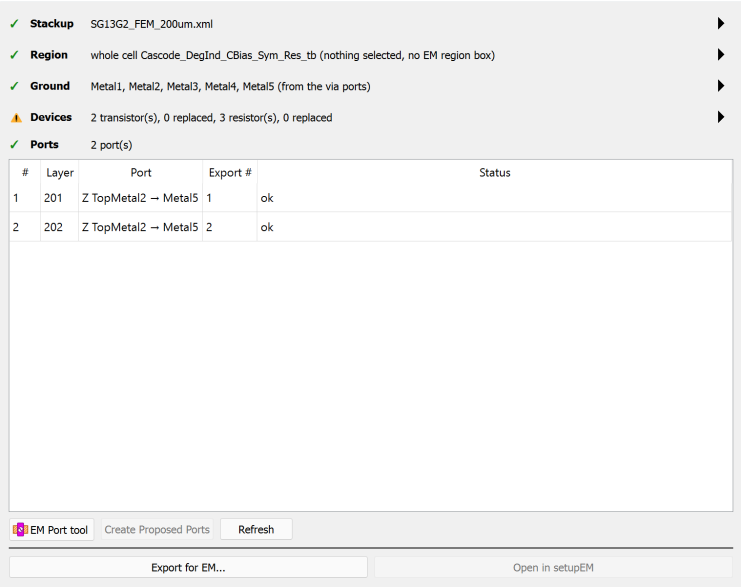


Figure 10: Two ports placed

EM > Port Overview shows the same as text, checked against the stackup:

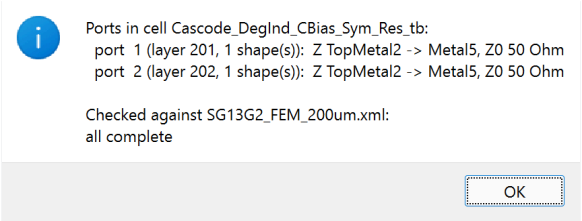


Figure 11: Port overview

What can go wrong:

- A port whose settings are incomplete is exported as a shape, but left out of the simulation. The panel marks it.
- A port that bridges the wrong two conductors measures something else than you intended. Chapter 7 checks the most common mistake, a port shorted by the layout.

7 Checks: ground and short circuits

In this chapter: the ground net the plug-in finds, and what its checks tell you.

Why it matters: a via port measures between a signal metal and ground. If the “ground” end is not connected to your ground, or both ends sit on the same net, the port measures a short circuit and the S-parameters are worthless. The plug-in traces the metal connections through the vias, so you see this before you simulate.

The plug-in finds the ground net by itself: of the nets your via ports reach, the one most ports reach is ground. You can also mark ground yourself.

1. Open the details of the **Ground** line.
2. Read the result: the ground metals, how many via ports reach ground, and how many other nets there are.
3. Press **Show in Layout** to see the ground net outlined in green.

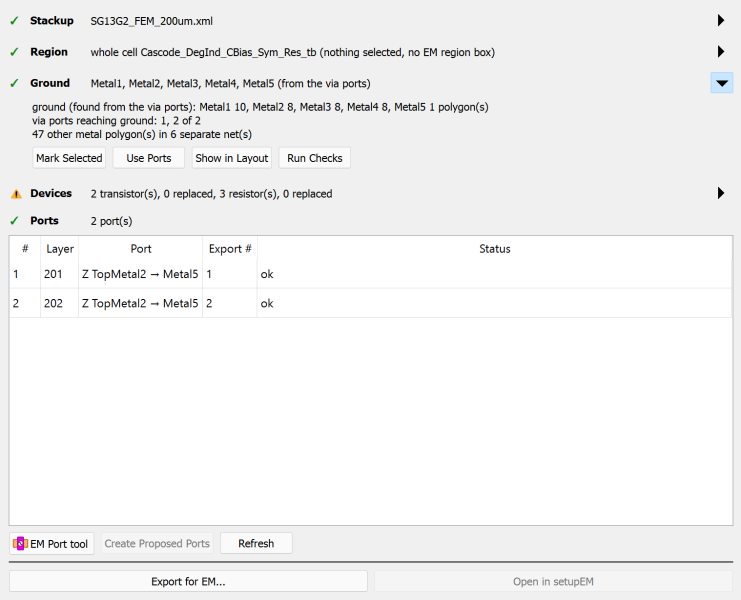


Figure 12: The ground check

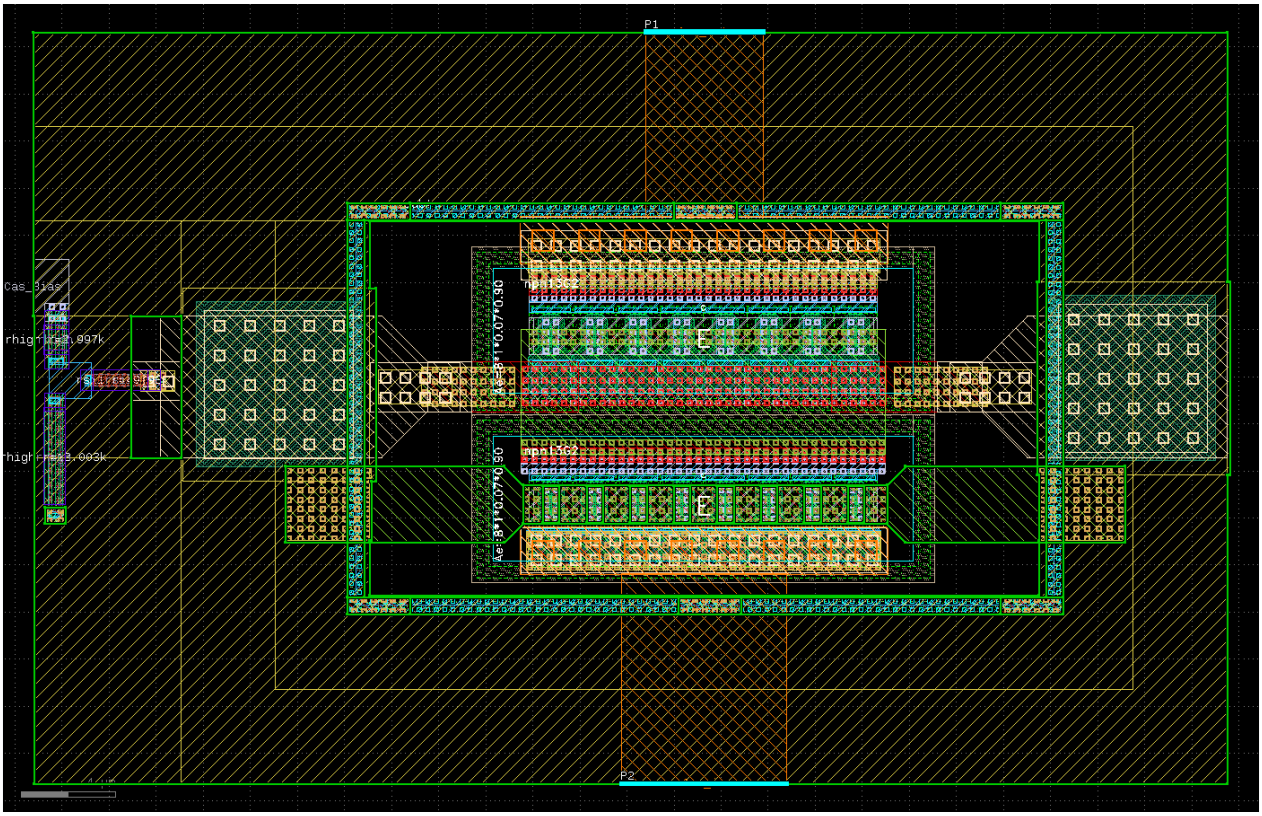


Figure 13: The ground net, outlined in green

The buttons:

- **Mark Selected**: select one or more metal shapes of your ground first, then press it. Use this if the automatic choice is wrong.
- **Use Ports**: forget the marks and use the automatic choice again.
- **Run Checks**: for large regions the checks do not run on every change; press it to run them.

What can go wrong:

- “short circuit: both ends of via port(s) … are on the same net”: that port connects two ends that are already connected by the layout. Check its from/to layers.
- “no ground: mark a ground shape or add via ports”: place the pad ports first, or mark ground.

8 Choosing the region to simulate

In this chapter: limiting the EM simulation to the part of the layout you need, and ports at wires you cut.

Why it matters: the EM simulation time grows quickly with the size of the region. Simulate the core with its wiring, not the whole chip. Where the region cuts through a wire, that wire needs a port, or its end simply floats in the simulation.

There are three ways to set the region:

- **Nothing selected, no box:** the whole cell.
- **A selection:** the export offers the selected shapes only, or everything inside their bounding box (chapter 10).
- **The EM region box:** a box on layer 800/0, saved with your layout. Select the shapes that define the region and press **EM > Mark EM Region from Selection** (or the button in the Region details), or draw the box yourself.

In the cascode, the region box is slightly smaller than the cell on the left. It cuts the Cas_Bias wire, the input of the bias divider. The plug-in sees the cut and proposes a port there:

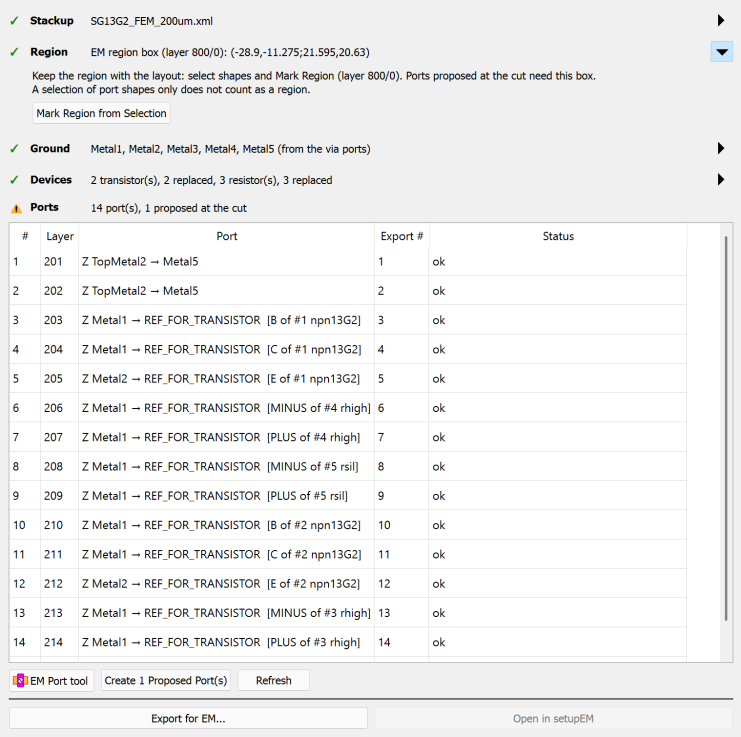


Figure 14: The region cuts the Cas_Bias wire: 1 port proposed

1. Look at the **Ports** line: “1 proposed at the cut”.
2. Press **Create Proposed Ports** (below the table). The plug-in places a via port at the cut, from the cut metal to the nearest ground metal.

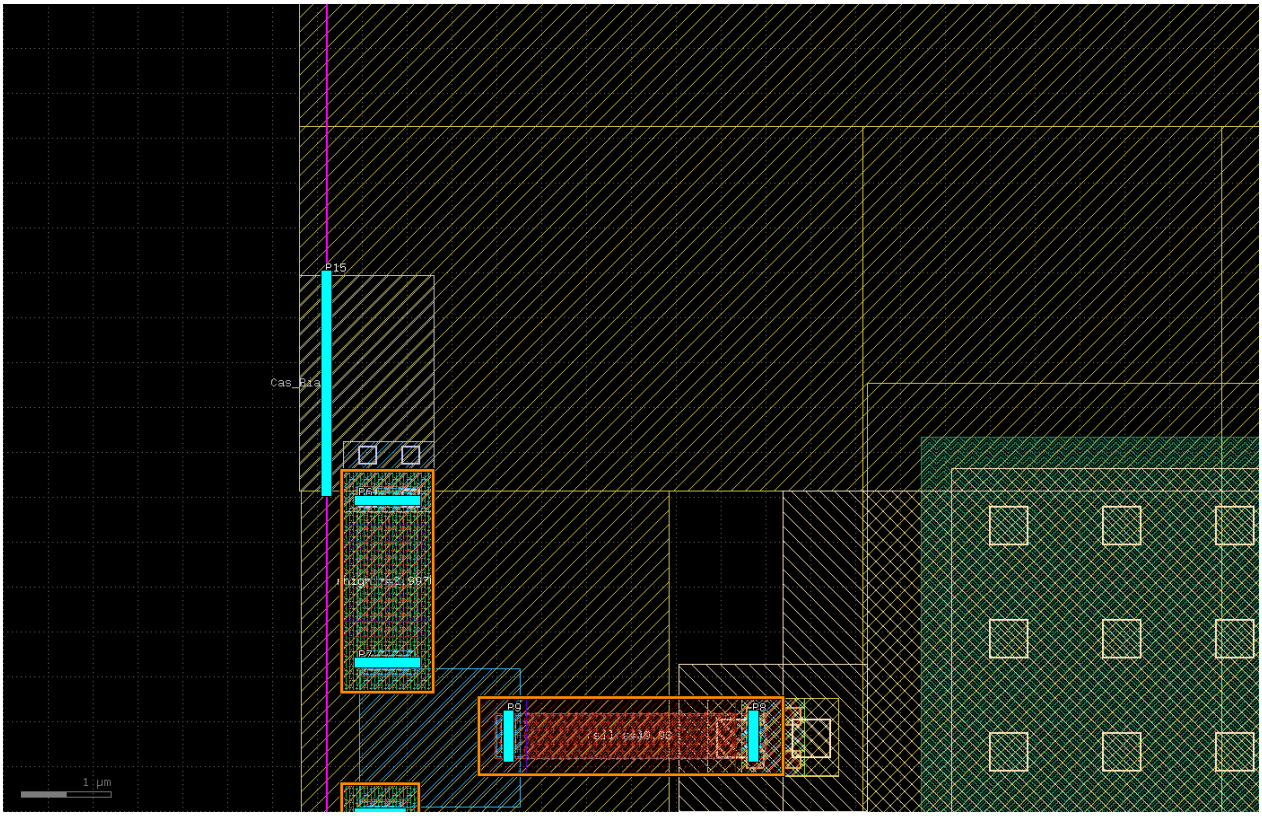


Figure 15: The new port 15 at the cut

The new port (P15 in the picture) lies on the region's edge, from the Cas_Bias wire on Metal2 to the ground plane on Metal5. The orange boxes are the transistors' and resistors' REF patches (chapter 9).

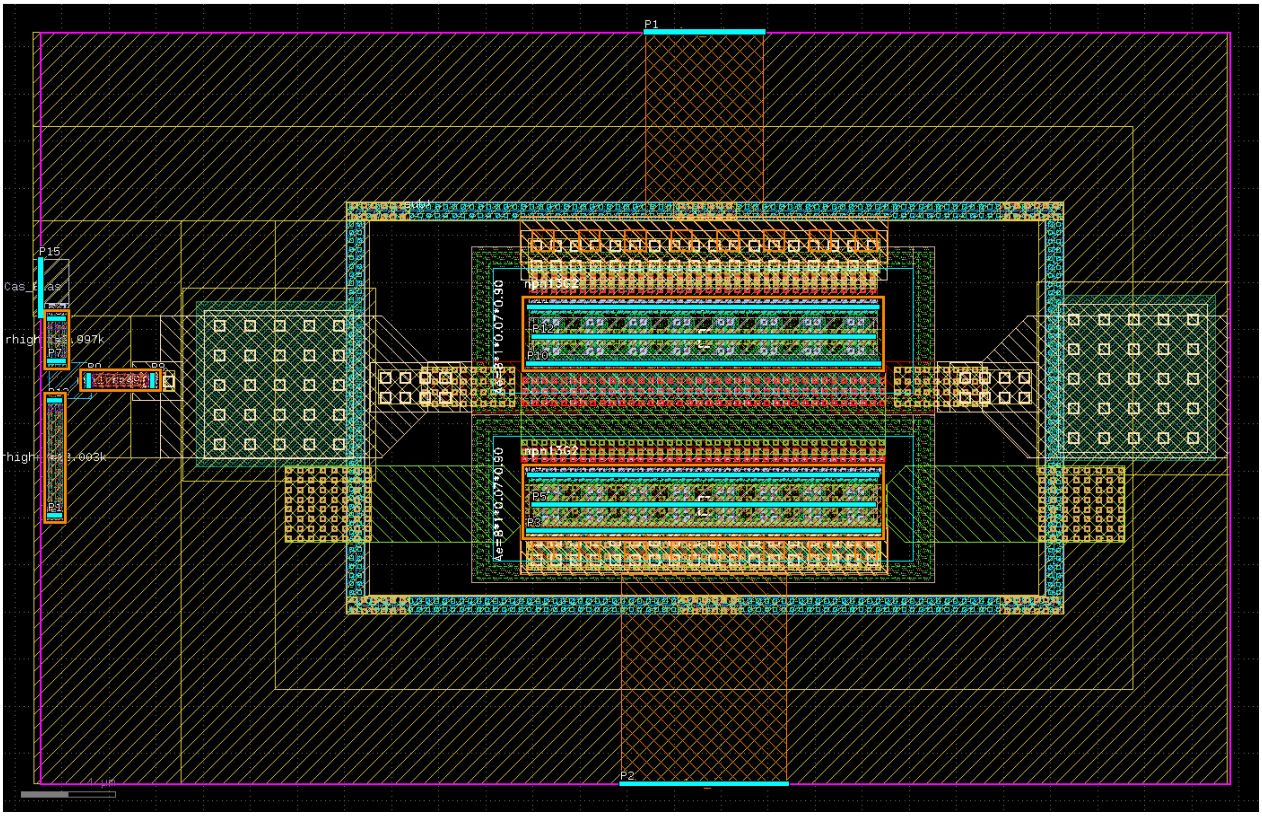


Figure 16: The whole region with all 15 ports

What can go wrong: “signal cut, but no ground metal at the cut: add a port by hand”: there is no ground metal near the cut to use as reference. Place that port yourself with the EM Port tool, or choose a region box that does not cut there.

9 Transistors and resistors

In this chapter: replacing the transistors and resistors by ports.

Why it matters: the EM simulation cannot simulate a transistor, and the PDK model already contains the device's own parasitics up to its pins. So the EM simulation should see the wiring up to the pins, and leave the device itself to the circuit simulator. Each transistor is replaced by one port per terminal; in the circuit, the PDK model is connected there again (chapter 12).

9.1 What happens

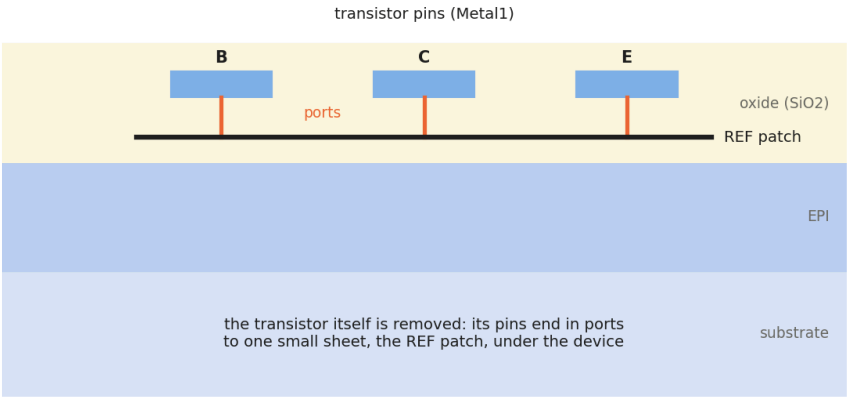


Figure 17: The REF patch: each transistor's ports end on one small sheet under the device

For each transistor, the plug-in:

- places a small sheet under the device, the **REF patch** (layer 300, REF_FOR_TRANSISTOR), just larger than the device's pins;
- places one via port per terminal, from the pin metal down to that sheet: base, collector and emitter for an npn, drain, gate, source (and bulk tie) for a MOSFET;
- at the export, removes the device's own shapes below the pins (and for the npn the small Metal1 islands under the emitter strip), because the PDK model covers them.

Each device has its own REF patch: the devices' ports must not be connected to each other through the patch.

Resistors (rsil, rppd, rhigh) are replaced the same way, with two ports each, and the PDK resistor model takes their place in the circuit.

9.2 Why resistors should be replaced too

The PDK does not compute a resistor's value from its drawn shape alone. Its pcells and SPICE models add two corrections:

- **A width offset.** The PDK uses an effective width that differs from the drawn one: rhigh 0.04 µm narrower, rsil and rppd 0.01 µm and 0.006 µm wider. This changes the width only, not the length.
- **An end resistance.** Where the silicided contact head meets the resistor body, each end adds a resistance per µm of width. It is not part of any drawn sheet.

An unreplaced resistor stays in the EM model as a sheet of the drawn size, with the nominal sheet resistance. The width offset is missing. Of the end resistance, the EM model has only what its contacts (Cont) contribute.

Both corrections are fixed amounts: a fixed width offset, and a fixed resistance at each end. They do not grow with the resistor. So the smaller the resistor, the larger their share, and the larger the error: physically small resistors are affected most. The width offset hurts narrow resistors, the end resistance hurts short ones.

How large is the error? Palace simulations of the PDK cells, as drawn, give:

Resistor (w × l)	PDK	EM model	Error
rhigh 0.655 µm × 1.245 µm (cascode, 3 kΩ)	2997 Ω	2599 Ω	−13 %
rppd 1 µm × 1.665 µm	500 Ω	442 Ω	−12 %
rhigh 0.5 µm × 4.29 µm (cascode, 13 kΩ)	13.0 kΩ	11.7 kΩ	−10 %
rhigh 2 µm × 1.325 µm	999 Ω	906 Ω	−9 %
rsil 0.5 µm × 2.33 µm (cascode, 50 Ω)	50.0 Ω	48.9 Ω	−2 %
rsil 2 µm × 13.06 µm	50.0 Ω	49.9 Ω	−0.2 %

rhigh and rppd come out about 10 % to 13 % low. rsil is close: its contacts in the EM model happen to make up for most of its end resistance.

Replaced, the resistor is the PDK model in the circuit: the correct value, with its temperature behaviour. That is why **include resistors** is ticked by default. If a resistor stays in the EM model, the **Devices** line shows orange and says so.

9.3 Step by step

1. Open the details of the **Devices** line. A table lists the transistors and resistors found: type, size or value, ports and status. Click a row to highlight the device in the layout, double-click to zoom to it. **#** is the device's number in Create Circuit's subcircuit (X1, X2, …, or R3 for a resistor taken from its label); the port table names it too, e.g. `[B of #1 npn13G2]`.

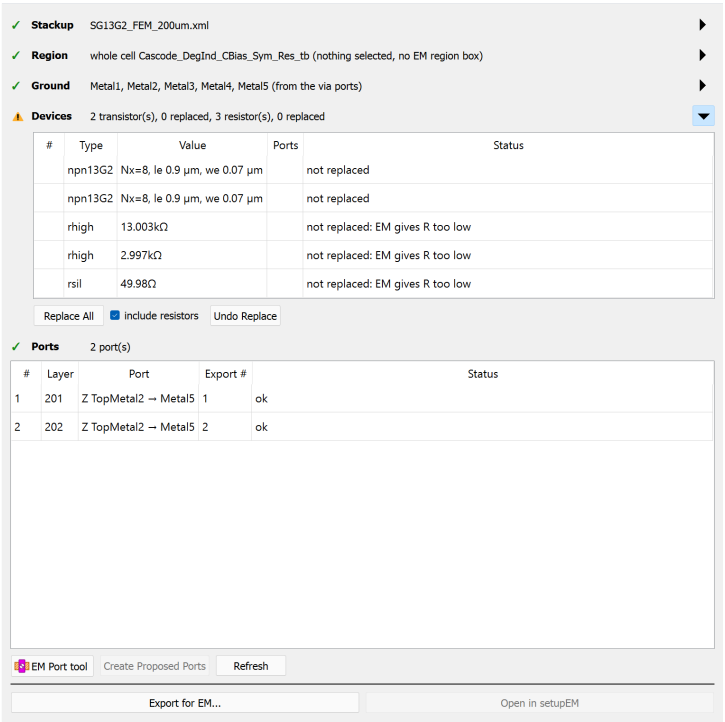


Figure 18: Devices found, none replaced yet

2. Leave **include resistors** ticked (the default). Untick it only if you want the resistors in the EM model, and accept that their values come out too low.
3. Press **Replace All**.

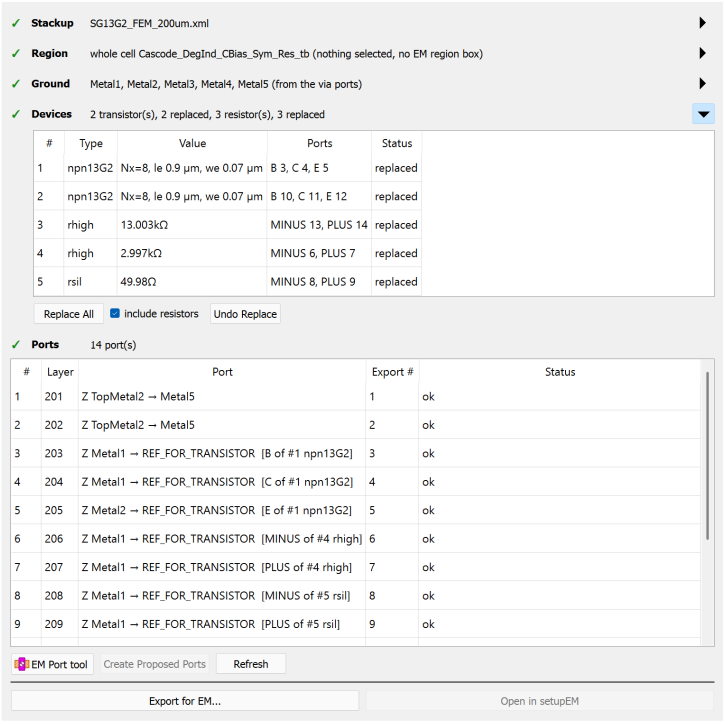


Figure 19: Two transistors and three resistors replaced

The ports appear in the table. In the layout, each transistor now has its REF patch (orange) and three ports (cyan):

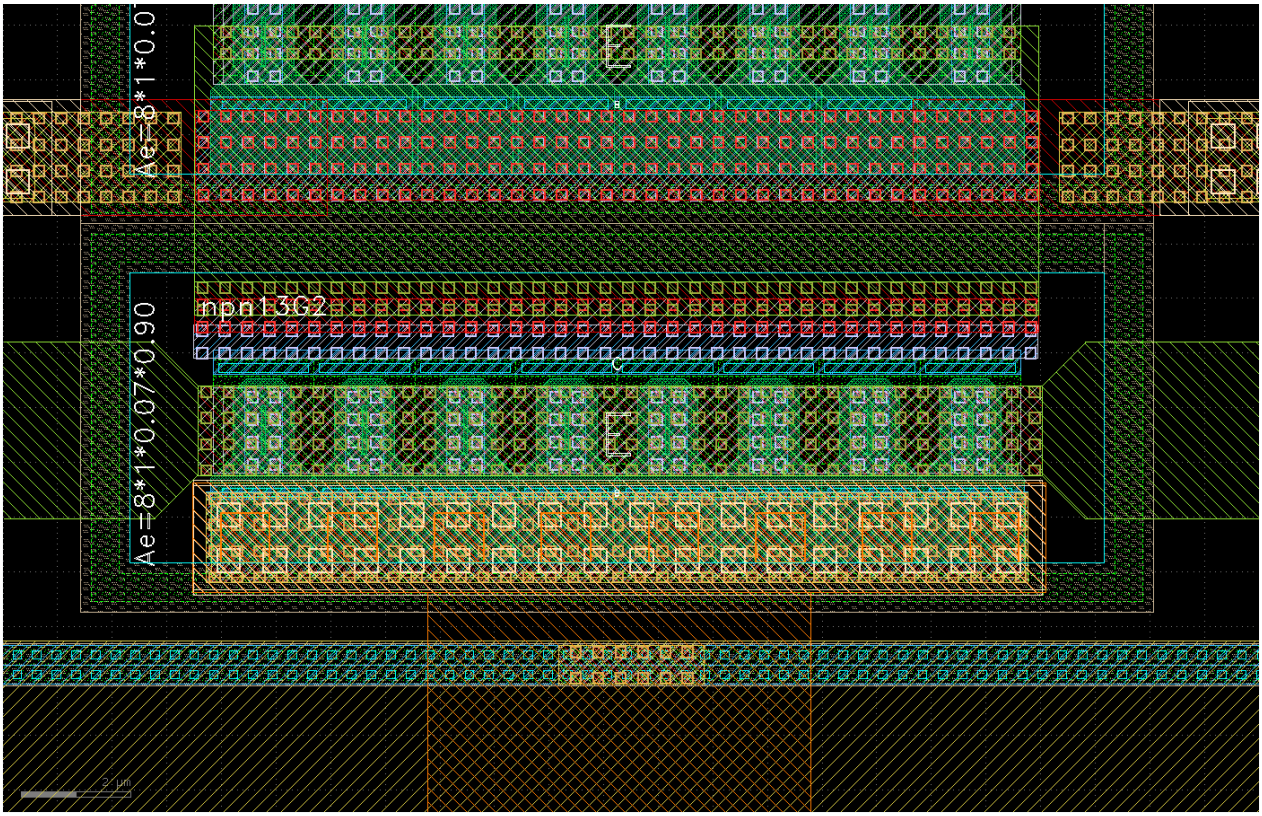


Figure 20: Q1 before

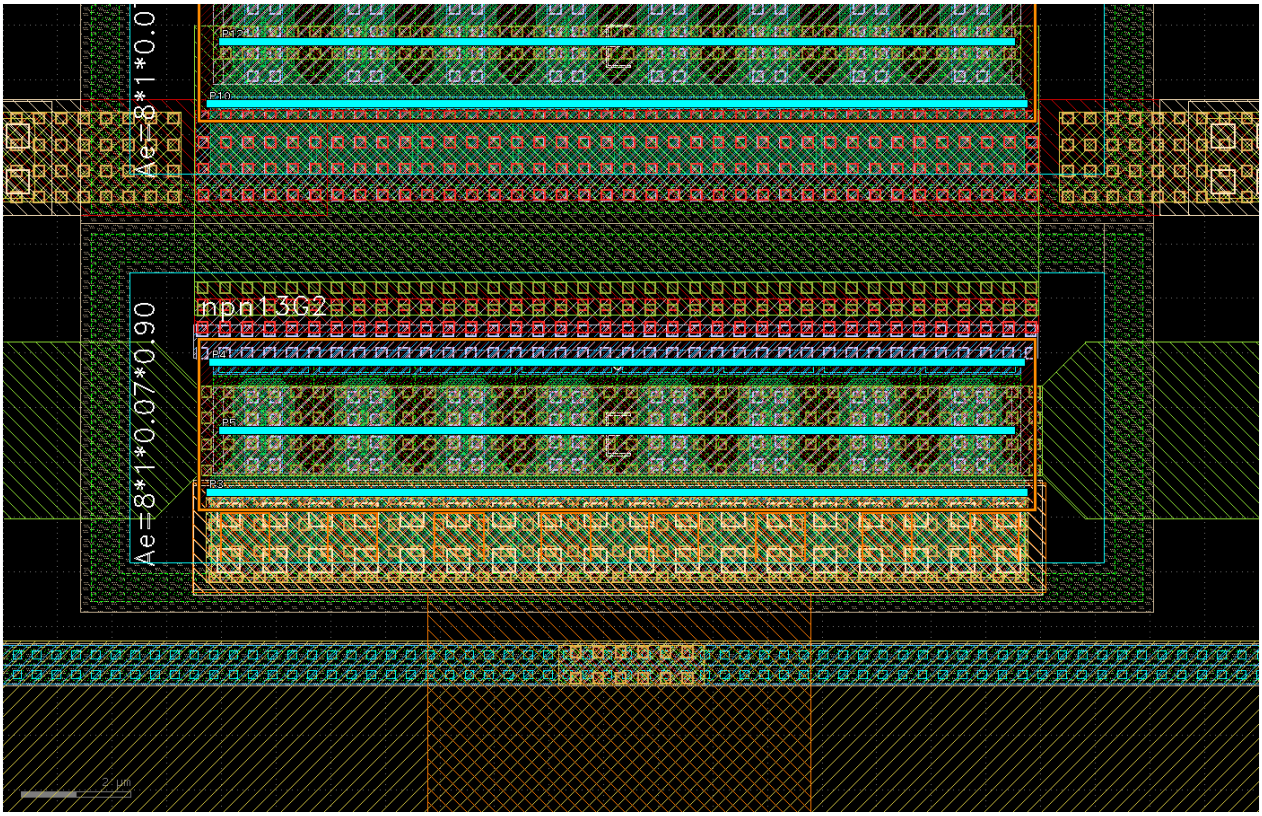


Figure 21: Q1 after: REF patch and ports P3 (base), P4 (collector), P5 (emitter)

Undo Replace removes all patches and ports made by Replace All.

What can go wrong:

- “no PDK devices (SG13_dev pcells) in the cell”: the IHP PDK is not loaded in KLayout, so the transistors are not recognized.
- “pair(s) of REF patches touch”: two devices sit so close that their patches overlap and would share one reference. The panel names them.
- “terminal(s) not connected”: a device terminal has no wiring in the region. Its port would measure an open end.

10 Export

In this chapter: writing the layout and the settings for the EM simulation.

Why it matters: the export writes a clean, flat layout with only the layers the EM simulation needs, plus the ports and a settings file for setupEM. It also writes a port map, which Create Circuit needs later.

When the panel shows green checks (or orange ones you have checked), press **Export for EM…**:

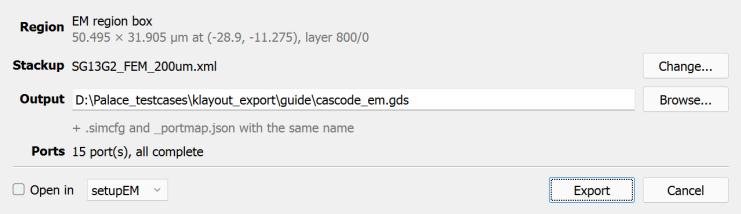


Figure 22: The export dialog

- **Region** and **Stackup**: as in the panel.
- **Output**: where to write the layout. The settings file for setupEM and EMStudio (`.simcfg`) and the port map are written next to it, with the same name.
- **Ports**: how many, and whether they are complete. A warning sign marks ports that need attention; the grey lines below say which and why.
- **Open in**: starts setupEM (chapter 11) or EMStudio with the result. The choice is remembered.

Press **Export**. You get three files, e.g. for `cascode_em.gds`:

File	Content
`cascode_em.gds`	the flat layout: stackup layers and ports only, devices removed below their pins
`cascode_em.simcfg`	setupEM's settings: layout, cell, stackup, ports
`cascode_em_portmap.json`	which port is what: pad, or which terminal of which device, and each port's net

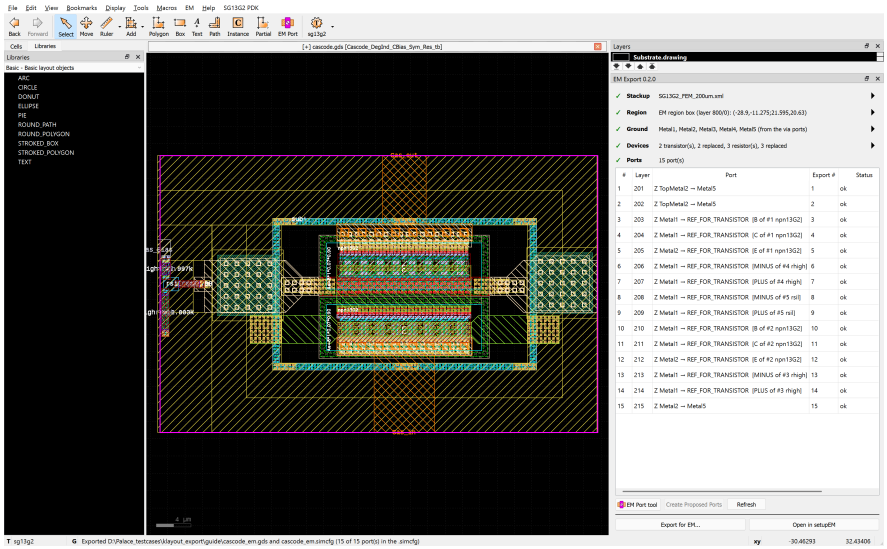


Figure 23: Ready: all checks green, exported

Good to know:

- **Port numbers:** the export numbers the ports 1, 2, 3, … without gaps. The port map links them back to the layout's ports. The panel's table shows both numbers.

- **Port direction:** the export sets each via port's direction (`Z` or `-Z`) so that all via ports have their signal metal on the same end. That is why port 15 (Metal2 below its ground on Metal5) comes out as `-Z` in setupEM.
- **Exporting again** keeps everything you set in setupEM (frequencies, mesh, …) and only updates layout, stackup and ports.
- **Open in setupEM** (or **Open in EMStudio**) in the panel opens the last export again, without exporting. The other program is in the EM menu.

What can go wrong: you export again after a layout change, but run the old model script (`.py`) from setupEM. The script reads the new layout under the same file name, with its old port list. Port numbers can shift when a port is added or removed, so the old script's port 3 may now be another port, and the S-parameters no longer belong to the port map. The export checks the model scripts next to the layout (and in setupEM's output folder) and warns about each one whose ports differ. Then open the export in setupEM: it asks whether to **Continue with the script**. Choose that, and setupEM takes the ports, layout file, stackup and cell from the export and keeps your other settings (frequencies, mesh, …) and any code you added to the script. Create Model writes the script. An EMStudio model next to the layout is updated by EMStudio itself when the export opens it there, see below.

11 setupEM and the Palace run

In this chapter: the EM simulation settings, and the one setting that matters most for the circuit.

Why it matters: for a circuit with bias, the EM result needs a good value at 0 Hz: the DC connections of your wiring. gds2palace provides that only if the frequency sweep starts at 0.

setupEM opens with your exported model. The **Input Files** tab shows the layout, cell and stackup from the export:

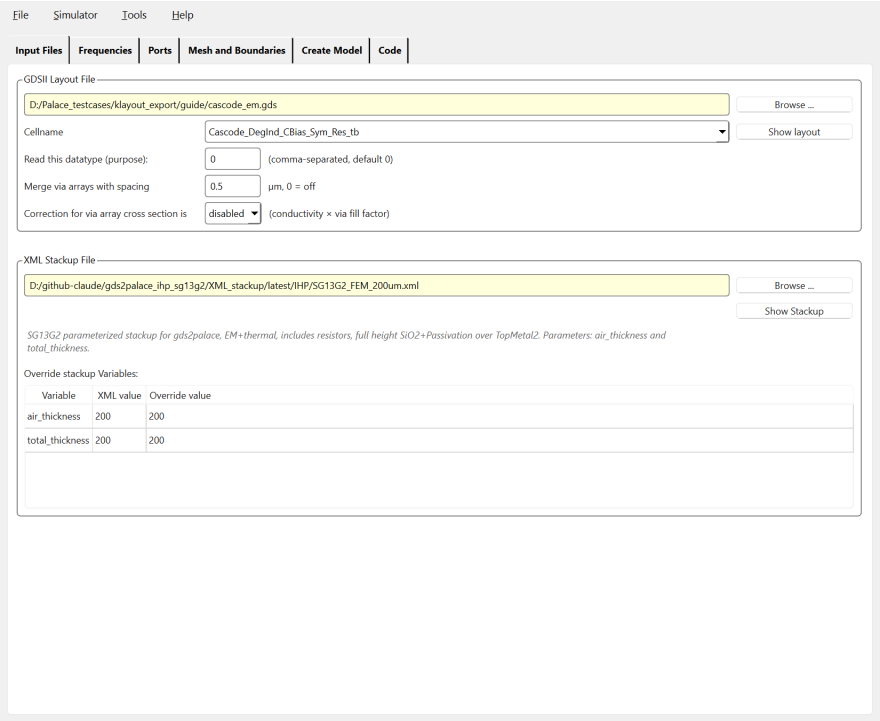


Figure 24: setupEM: input files

The **Ports** tab shows the exported ports:

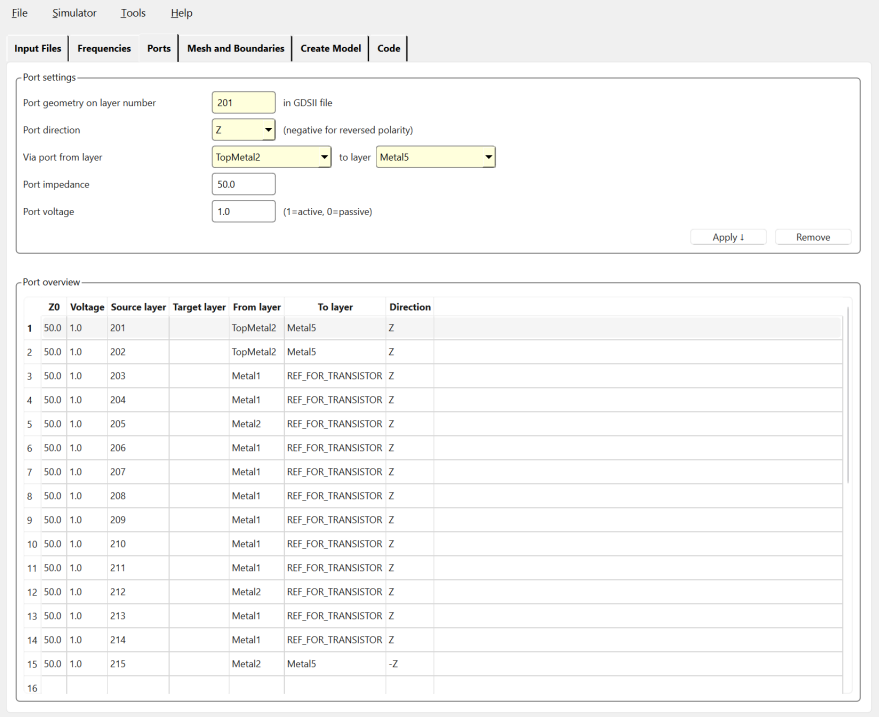


Figure 25: setupEM: ports

11.1 Frequencies: start at 0

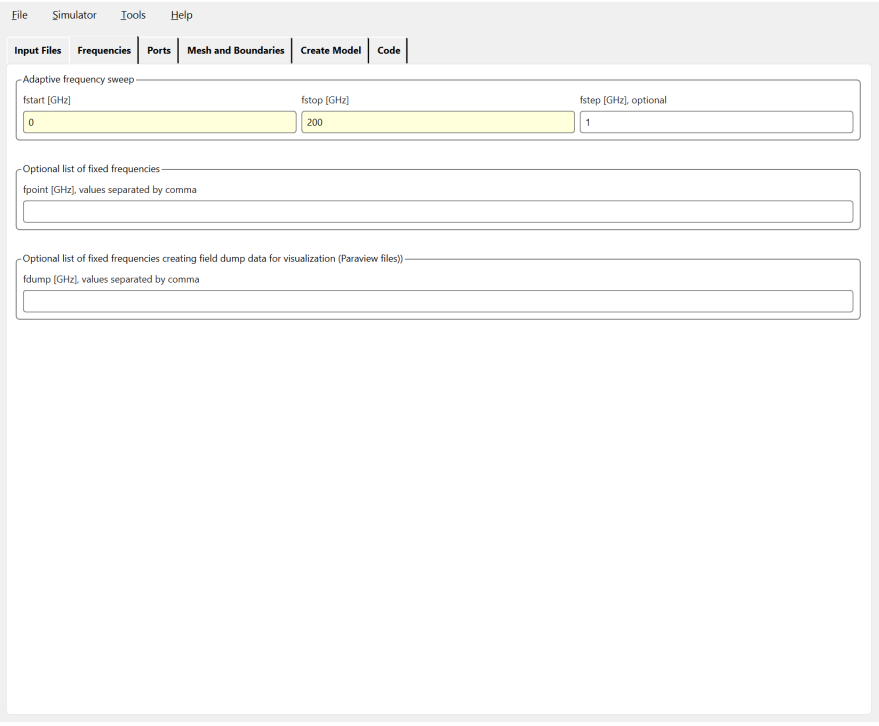


Figure 26: setupEM: frequencies, from 0 to 200 GHz in 1 GHz steps

1. Set **fstart** to **0**.
2. Set **fstop** to the highest frequency you need (200 GHz for the cascode), and **fstep** (1 GHz).

With fstart = 0, gds2palace simulates 10 MHz and 20 MHz in addition, and writes a second result file, `<model>_dc.sNp`, with a 0 Hz point extrapolated from them. That is the file for chapter 12.

11.2 Mesh and model

Mesh and boundaries are explained in the setupEM and gds2palace documentation. For a first run, the defaults work.

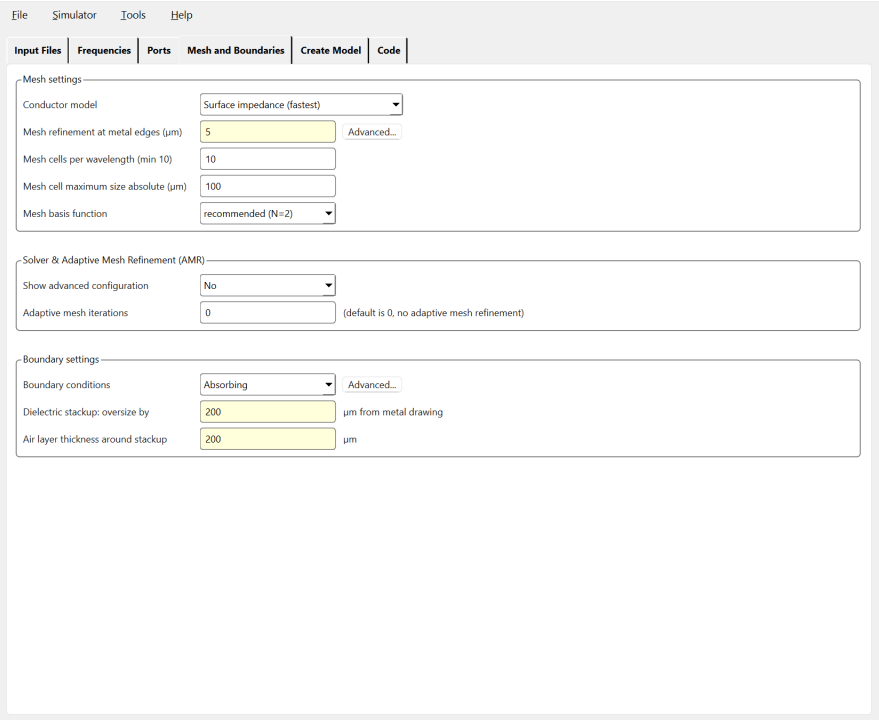


Figure 27: setupEM: mesh and boundaries

On the **Create Model** tab, create the Palace model, then run Palace (on Linux, e.g. with the `run_palace` script of gds2palace). After the run, `combine_snp` writes the S-parameter files, for the cascode `<model>.s15p` and `<model>_dc.s15p`.

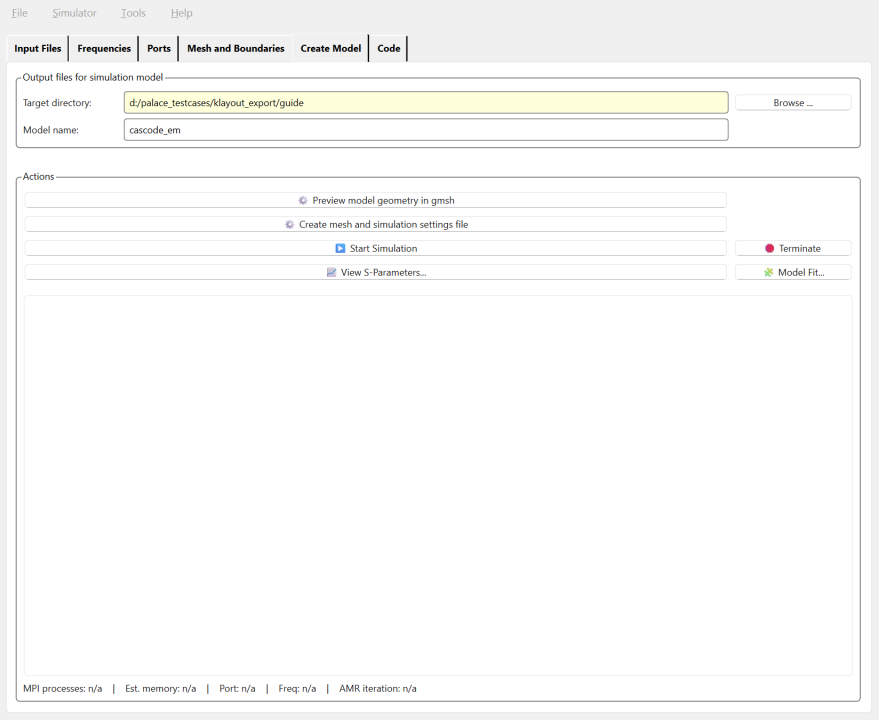


Figure 28: setupEM: create model

What can go wrong: a sweep that starts at 1 GHz or higher gives a 0 Hz point that is extrapolated too far. Its DC paths are not accurate enough for a bias network with kOhm resistors. Create Circuit then protects the bias with DC blocks (chapter 12), but the cleaner way is to start at 0.

11.3 EMStudio instead of setupEM

EMStudio does the same job in one window: model settings, the simulation run and the results. Choose **EMStudio** next to **Open in**. EMStudio then reads the export the way setupEM does, and asks whether the model is for **Palace** or **OpenEMS**.

- **The first time**, EMStudio makes a new model for the simulator you chose: layout, cell, stackup and ports. Set the frequencies there, starting at **0 GHz** as above, and save. The model is saved as `<name>.py` next to the layout, e.g. `cascode_em.py`.
- **After a layout change**, export again. EMStudio finds `cascode_em.py` and updates only its ports, layout, stackup and cell. Your frequencies and mesh stay. Save to keep the update.
- **The stackup always comes from the export.** A stackup you picked in EMStudio is replaced on the next export, so choose it in the export dialog.
- **The first import fixes the simulator.** EMStudio asks only once. For the other simulator, rename or delete `cascode_em.py` and export again; you get a new model and set its frequencies again.
- **Keep the model where EMStudio put it.** If you save it elsewhere or under another name, the next export does not find it and makes a new model.
- The port directions arrive as exported, `-Z` included, so the circuit wrapper fits.

For OpenEMS, prepare the export differently:

- **Stackup:** choose an OpenEMS stackup (from gds2openEMS), never a Palace one. They model some layers, e.g. the MIM dielectric, differently.
- **Devices:** don't replace them. Device ports end on a reference layer that only the Palace stackups have; for OpenEMS, simulate the layout without device replacement.

Nothing stops you from mixing these up, so check both before the export.

You need an EMStudio whose **File > Import setupEM Config** asks for Palace or OpenEMS (IHP-GmbH/EMStudio PR #53). An older build has the menu too, but imports every export as a Palace model. If the plug-in cannot find EMStudio, it asks once where the program is.

12 Back to the circuit: Create Circuit

In this chapter: turning the EM result into a circuit with your transistors, for your circuit simulator.

Why it matters: this is where the EM work pays off: you simulate the amplifier with its real wiring, with the PDK transistor models and their bias.

12.1 Creating the circuit

EM > Create Circuit (EM Result with Devices)…:

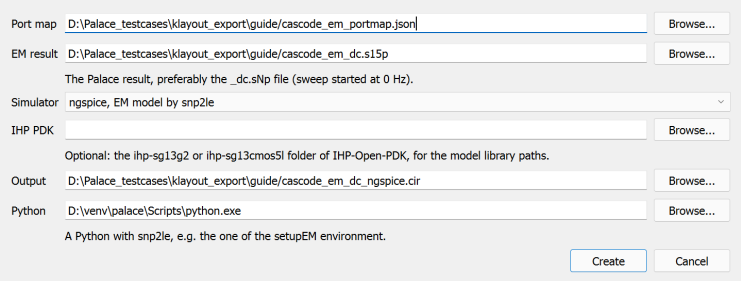


Figure 29: The Create Circuit dialog

1. **Port map**: the export's `_portmap.json` (the last export of this layout is filled in).
2. **EM result**: the Palace result, preferably the `_dc.sNp` file.
3. **Simulator**: see below.
4. **IHP PDK** (optional): the `ihp-sg13g2` or `ihp-sg13cmos5l` folder of IHP-Open-PDK. Then the model library lines in the circuit file point to it.
5. **Output**: the circuit file (proposed from the result's name).
6. **Python**: a Python with snp2le, by default the one of the setupEM environment.
7. Press **Create**.



Figure 30: Choose the simulator

Choice	The EM result goes in as	Use it for
ngspice, EM model by snp2le	an equivalent circuit (fitted by snp2le)	DC, AC, transient, noise
VACASK, EM model by snp2le	an equivalent circuit (VACASK format)	as ngspice, plus harmonic balance
Xyce, EM model by snp2le	an equivalent circuit	DC, AC, transient, harmonic balance
Xyce, the .sNp file	the S-parameter file itself (YLIN element)	harmonic balance only: Xyce treats it as open in AC and DC
Spectre, the .sNp file	the S-parameter file itself (nport element)	all analyses; add your PDK's model includes

The result shows what was written:

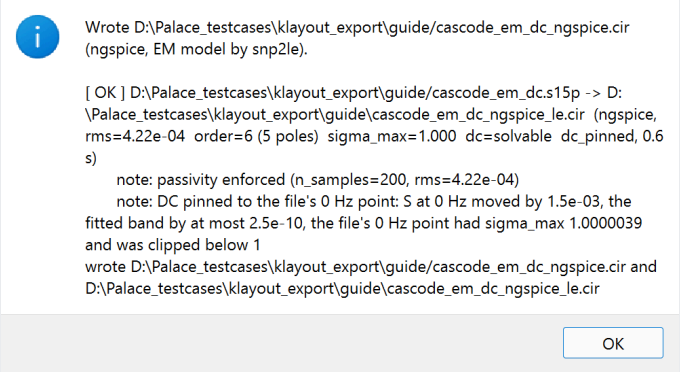


Figure 31: Done: the circuit and its equivalent circuit

For the snp2le choices you get two files: the circuit (`..._ngspice.cir`) and the EM model's equivalent circuit (`..._le.cir`), which the circuit includes.

12.2 What is in the circuit

The circuit is one subcircuit. Its pins are the ports that belong to no device: the pads and the bias input. Inside:

- **the EM model**, with one pin per port;
- **the PDK transistor models**, each terminal connected directly to its port's pin (npn13G2 with Nx, le, we from your layout);
- **the resistors**: PDK models, or ideal resistors of the label value for static resistor cells;
- **1 mH inductors** between the ports of each net, and to ground for the ground net.

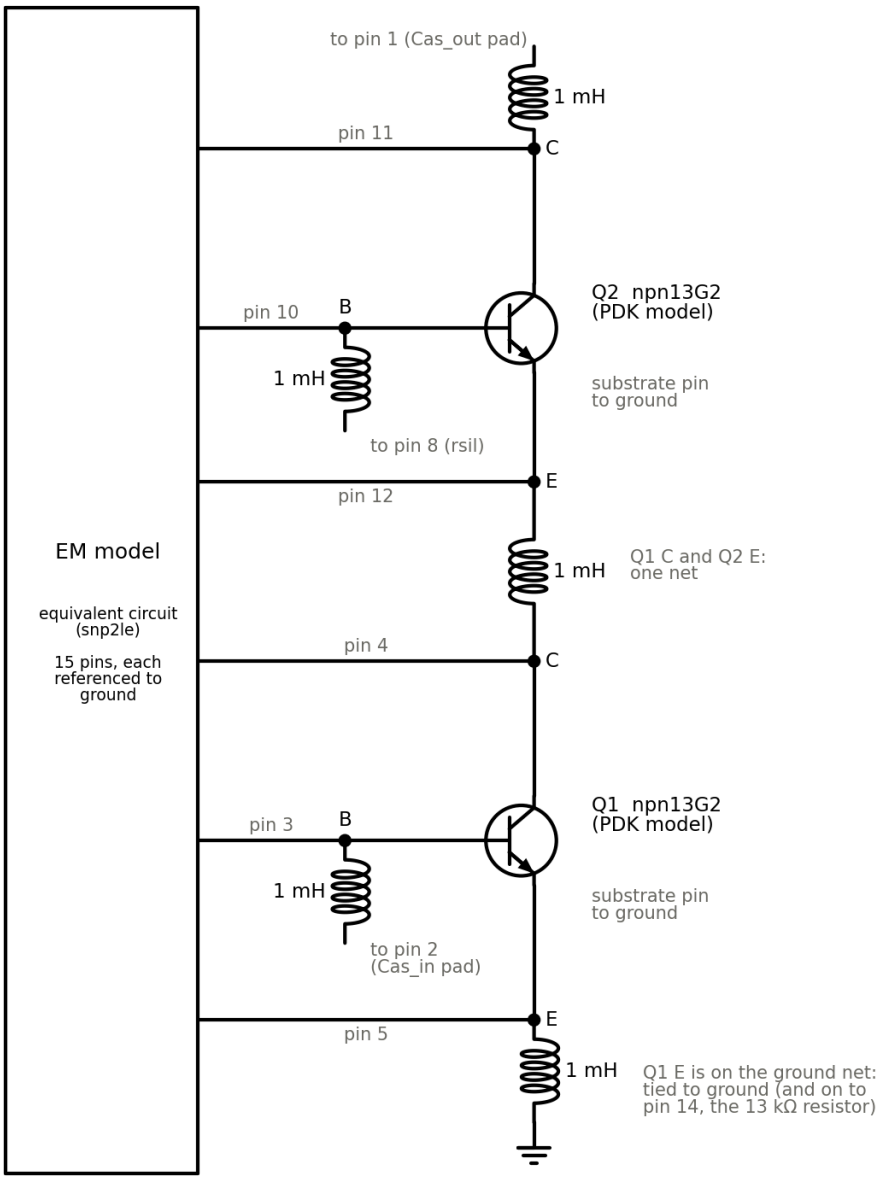


Figure 32: Around the transistors: device pins on the EM pins, and the 1 mH ties

Why the 1 mH inductors? A port measures a voltage between a device pin and its REF patch. The patch has no DC connection to anything, so the EM result alone does not tell the circuit simulator at which DC potential a transistor sits. The inductors tie each terminal to the rest of its net: at DC they are shorts and carry the bias, at RF (GHz) they are open and leave the signal to the EM model. The port map tells which ports are on the same net.

The full circuit with the testbench, the layout's wiring drawn as dashed lines inside the EM model:

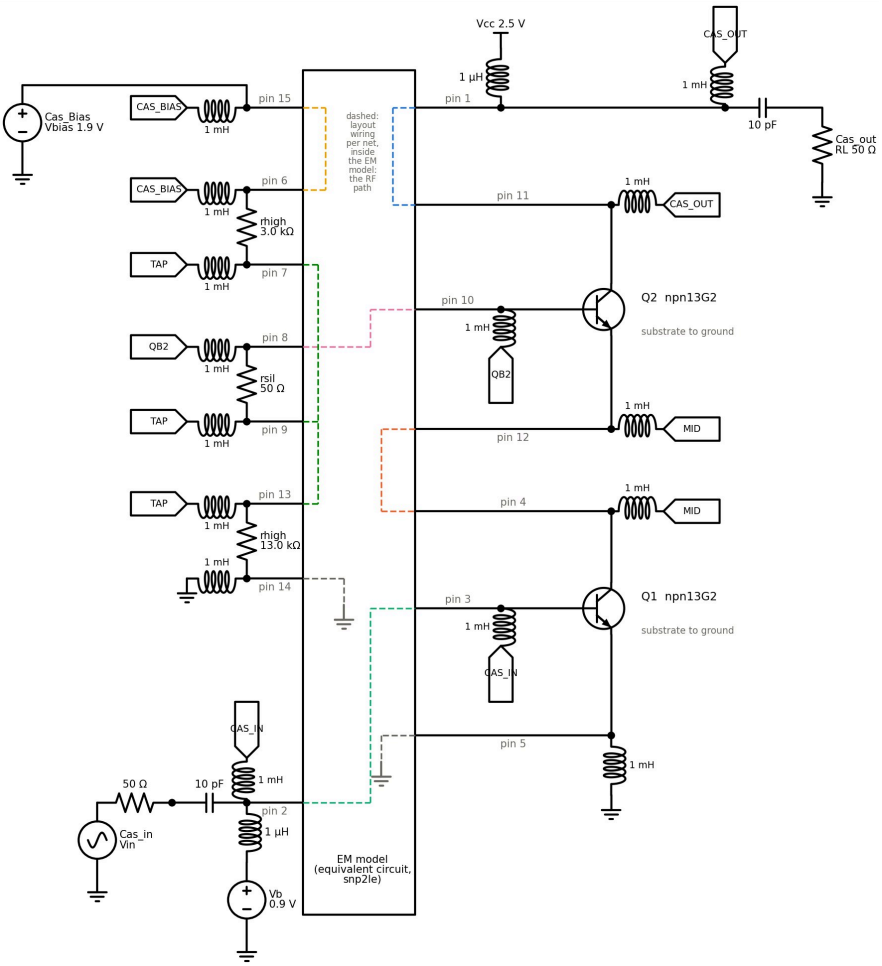


Figure 33: The cascode as simulated

12.3 Using it in your testbench

ngspice example (`--pdk` given, so the model library lines are active):

```
.include cascode_em_dc_ngspice.cir
Vcc  vcc 0 2.5
Lc   vcc port1 1u          ; bias tee at the output
Cout port1 out 10p
Rl   out 0 50
Vb   vb 0 0.9
Lb   vb port2 1u           ; bias tee at the input
Vin  src 0 dc 0 ac 1
Rs   src in 50
Cin  in port2 10p
Vbias port15 0 1.9         ; Cas_Bias
X1 port1 port2 port15 cascode_em_dc_ngspice
```

- **VACASK:** include the `.inc` file. VACASK finds the PDK's models through the PDK's `.vacaskrc.toml`: copy it next to your testbench.
- **Spectre:** the PDK include lines are written as comments: replace them with the includes of your SG13G2 PDK installation.

What can go wrong:

- "no 0 Hz point in the .sNp: the EM model is connected through 1 uF DC blocks": your result has no DC point (chapter 11). The circuit still works, the DC blocks keep the EM model out of the bias.
- "the port map has no signal layers (older export)": export again with the current plug-in.

- snp2le not found: choose the Python of the environment where snp2le is installed.

13 The cascode example: results

In this chapter: what the flow gives for the example, and how much each modelling choice matters.

Why it matters: it shows what to expect, and which numbers to check in your own design.

The exported cascode, with all 15 ports:

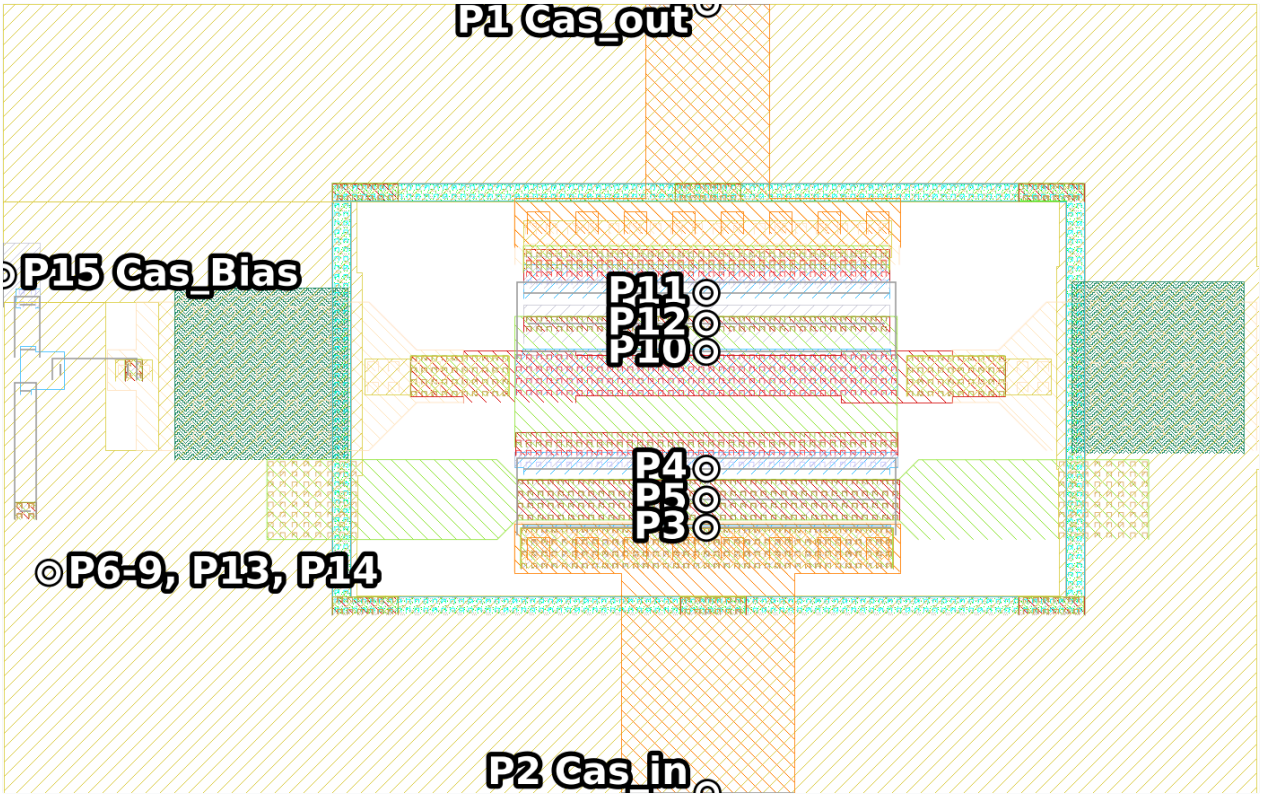


Figure 34: The exported cascode with its ports

The EM model as a circuit: snp2le's equivalent circuit reproduces the Palace result within 0.003 in S, for all 225 S-parameters from 1 to 200 GHz:

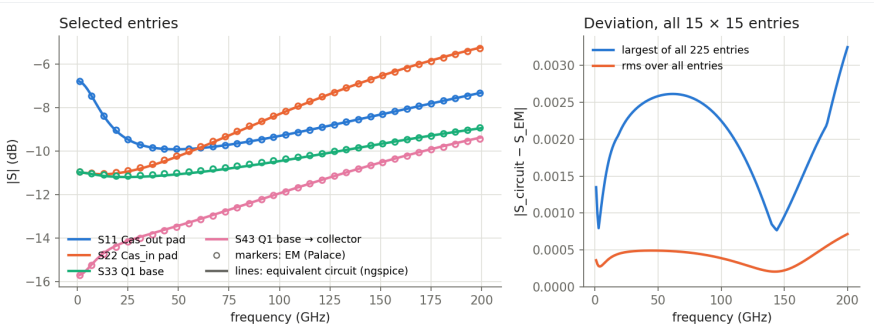


Figure 35: The equivalent circuit against the EM result

The amplifier: with Vcc = 2.5 V, Vb = 0.9 V and Cas_Bias = 1.9 V, the cascode draws Icc = 11.3 mA, and S21 is 21.6 dB at 10 GHz, 7.9 dB at 100 GHz and −1.0 dB at 200 GHz. The core is unmatched: it has no matching networks.

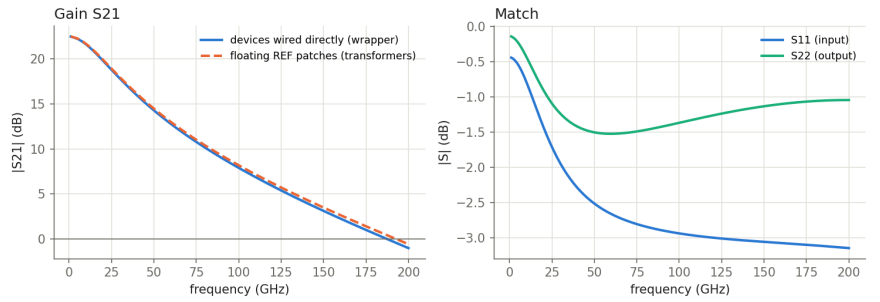


Figure 36: Gain and match of the cascode

The simulators agree: VACASK gives the same result as ngspice. Xyce gives the same operating point, and S21 0.5 dB higher at 100 GHz, which comes from Xyce's version of the transistor model, not from the EM model.

The cost of the simple wiring: connecting the transistors directly to the EM model ties their REF patches to ground, which adds a few fF. In the cascode that lowers S21 by 0.06 dB at 10 GHz and 0.4 dB at 200 GHz.

14 Thermal simulation

In this chapter: the temperature of each transistor and resistor, from the same layout, and back to your circuit.

Why it matters: a transistor that dissipates power heats itself and its neighbours. Its temperature changes its current, its gain and its lifetime. The self-heating in a compact model knows only the device itself: not the neighbours, not the wiring that carries heat away, not the substrate and package below. A thermal simulation of the layout includes all of that.

The thermal flow starts from the same layout and region as the EM flow. It needs no ports, and the devices stay as they are:

1. **In KLayout**, the EM panel's **Thermal** tab lists one heat source per transistor and resistor. **Export for Thermal…** writes the model for setupThermal.
2. **Your circuit simulation** gives the power of each device.
3. **In setupThermal**, **Load powers…** sets these powers on the heat sources. You choose where the temperature is fixed, and Elmer computes the temperatures.
4. **The temperature of each device** comes back as a table with its schematic instance name, for your circuit simulation.

14.1 The Thermal tab

Switch the EM panel to **Thermal** at the top:

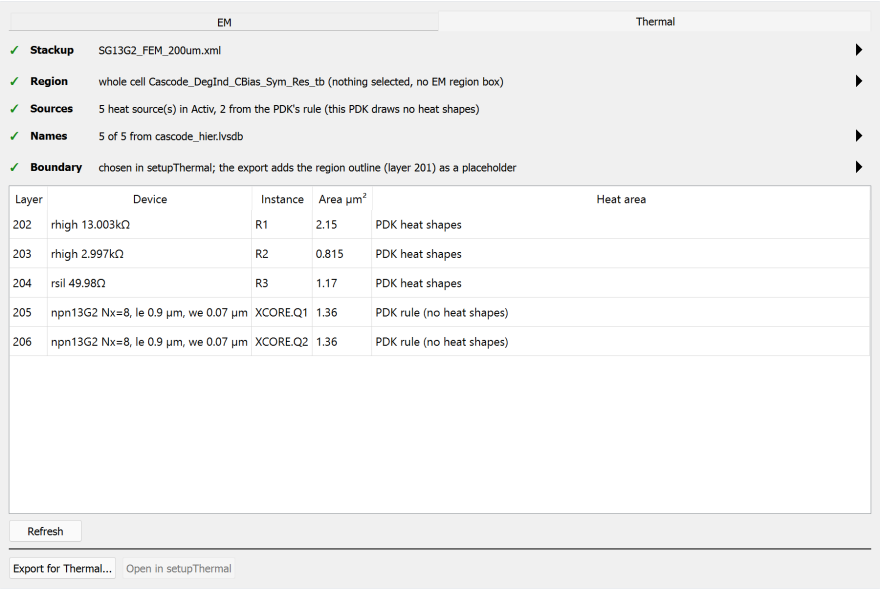


Figure 37: The EM panel in Thermal mode, with the heat sources of the cascode

- **Stackup** and **Region**: as for EM (chapters 5 and 8). The stackup needs thermal data: the gds2palace stackups for EM and thermal, e.g. `SG13G2_FEM_200um.xml`, have it.
- **Sources**: the heat sources, one per device in the region, with its marker layer, device, schematic name, heat area and where the heat area comes from.
- **Names**: the schematic instance names, from your LVS run (see below). Optional.
- **Boundary**: the fixed temperatures of the last thermal export. You choose them in setupThermal.

The EM steps (ground, devices, ports) are hidden. A thermal model needs none of them.

14.2 Heat sources

A transistor generates its heat in a small volume: at the emitter of an npn, under the gate of a MOSFET. A resistor heats along its body. The temperature is highest there and falls steeply around it. So the heat source must have the size of that volume, not the size of the whole device. A larger area spreads the heat and gives too low a temperature.

The IHP PDK draws these areas itself, on **HeatTrans (51/0)** and **HeatRes (52/0)**: one rectangle per emitter or gate finger, and the body of each resistor. The export takes them. PDK cells without them (older PDK versions, or static cells) get the same rectangles from the device's own layers, as the PDK does it: each npn emitter window (EmWind, 33/0) grown by 0.05 µm, each MOS gate finger, each resistor body. The **Heat area** column says which one was used. Only a device with neither gets its bounding box, with a warning.

Each heat source:

- goes on a marker layer of its own, from 201 upward. Layers that the stackup or your own markers use are skipped;
- sits in the height of **Activ**, the active silicon at the top of the substrate;
- gets one power, which Elmer spreads over the device's fingers by their area.

The metal, contacts and vias of the devices stay in the model, since they carry heat away.

Good to know: the heat areas are small and lie under all the metal. To see them in KLayout, show all hierarchy levels (*), hide all layers, and switch on HeatTrans and HeatRes (or EmWind, for an npn without HeatTrans). In the exported layout they are on the marker layers.

14.3 Instance names from LVS

The powers come from your circuit simulation, per schematic instance: `Q1`, or `Xcore.Q1` for a device in a subcircuit. To find each device's instance, the plug-in reads the database of a KLayout LVS run of this layout against your schematic. IHP's LVS deck writes it (`.lvsdb`).

Choose the file in the **Names** step (or in the export dialog). Each heat source then gets its instance name. LVS may have been run on the exported cell, or on a cell that contains it once. A heat source without a matching LVS device gets no name, and the Names step lists it.

Without LVS, the export still works. The heat sources then have only their layout names, e.g. `npn13G2@-7230,590`, and the powers file must use those.

14.4 Export for Thermal

Press **Export for Thermal…**:

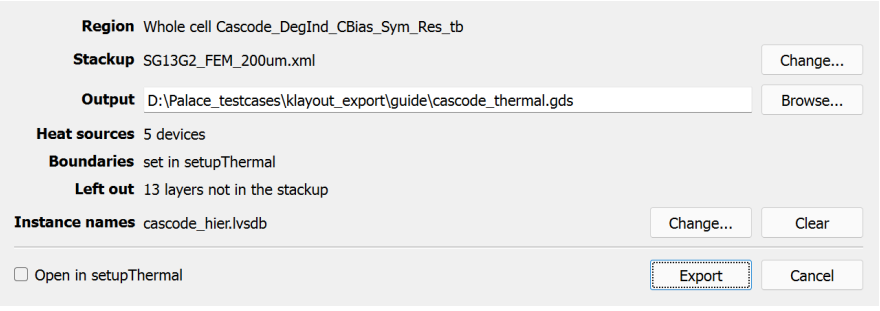


Figure 38: The thermal export dialog

- **Heat sources**: how many, and which of them have no heat area of their own.
- **Boundaries**: your own boundaries in the layout, if any.
- **Left out**: layers with shapes in the region that are not in the stackup.
- **Instance names**: the LVS database.

You get three files, e.g. for `cascode_thermal.gds`:

File	Content
`cascode_thermal.gds`	the flat layout: stackup layers with the devices as drawn, the heat sources and the region outline on marker layers
`cascode_thermal.tsimcfg`	setupThermal's settings: layout, cell, stackup, one heat source per device (0 W) and a placeholder boundary
`cascode_thermal_thermalmap.json`	which device and instance each heat source layer belongs to

Exporting again keeps everything you set in setupThermal: mesh, boundaries, and the power and name of every device still in the region.

14.5 Boundaries

Heat must leave the model somewhere. Usually that is the backside of the chip, on a heat sink or a carrier. Which face is right depends on the assembly, so the export does not choose it. It writes the region outline on the first marker layer, with a placeholder: a fixed temperature of 300 K without a target layer.

setupThermal lists it as **region outline: choose the target layer**, and does not build the model until you do. The target layer is the height of the fixed temperature: e.g. **BACKSIDEGND** for the substrate backside, if your stackup has it. Set the temperature too, e.g. the heat sink's.

The side faces of the model are adiabatic: no heat leaves there. Choose the region with enough substrate around the devices, so that this does not matter.

14.6 Your own heat sources and boundaries

You can also draw your own markers, e.g. a pad area with a fixed temperature, or the heat source of a block without PDK devices. The export keeps them on their layers. It recognizes them in two ways:

- the layers your own setupThermal file for this layout uses (a `.tsimcfg` next to the layout, with this GDS file);
- shapes on 201 … 299 that are not EM ports.

Their settings (power, temperature, target layer) go into the exported `.tsimcfg`.

EM ports and thermal markers in one layout: give them different layer numbers. EM ports are on 201 … 299. Put thermal markers e.g. on 401 … 499, which no stackup of gds2palace uses.

14.7 In setupThermal

Open in setupThermal in the panel opens the export. Then:

1. **Powers:** in the **Thermal Sources + Boundaries** tab, **Load powers…** reads a file from your circuit simulation, one line per device: instance name and power in W, separated by a comma, semicolon, tab or spaces.

   ```
   instance, power
   Xcore.Q1, 12.5m
   Xcore.Q2, 12.5m
   R1, 0.2m
   ```

   SPICE suffixes are fine (`12.5m`, `12.5mW`, `12.5e-3`), and so is a header line. setupThermal reports heat sources without a power and names that match none. Nothing is set to 0 W silently.
2. **Boundary:** choose the target layer of the region outline (see above).
3. **Mesh:** the mesh at the heat sources must be fine. An npn's emitter is 0.17 µm wide, and a mesh that is too coarse there, or grows too fast, gives too low a temperature. Set a fine value only where it is needed, with **Advanced…** next to **Mesh size at heat sources**: e.g. 0.125 µm for the npn sources and the default for the resistors. The setupEM guide explains the mesh settings (Mesh tab, setupThermal).
4. **Solver:** for a large mesh, choose the iterative solver. The direct solver needs much more memory: on a laptop it failed at about a million elements.
5. **Create Model** and run.

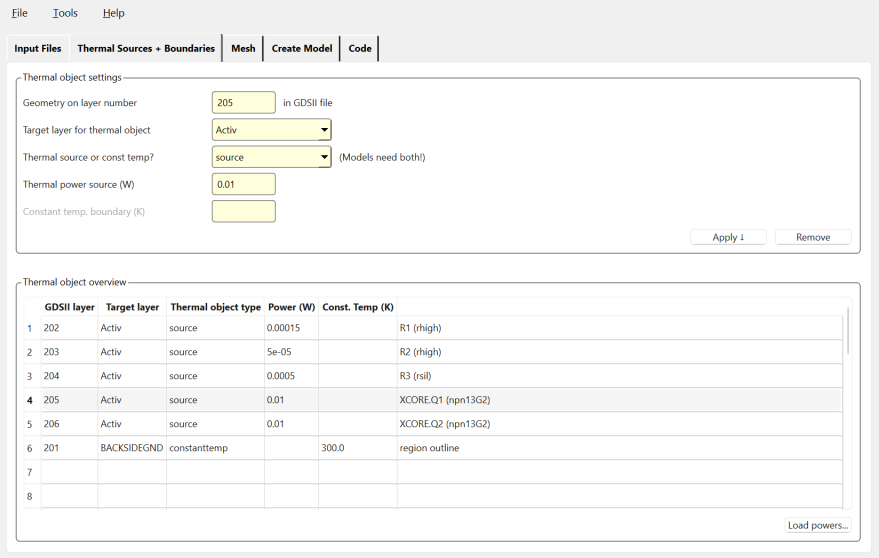


Figure 39: setupThermal after Load powers…, with the boundary on BACKSIDEGND (example powers)

14.8 Temperatures back to the circuit

After the run, setupThermal lists the temperature of each heat source in the log, and writes the same table to `<model>_temperatures.csv` next to the model script. The command line does the same without the GUI: `python -m setupEM.thermal_results <model>.tsimcfg`.

Column	Meaning
`instance`	the schematic instance name (from LVS)
`device`, `pcell`, `layer`	the layout device and its heat source layer
`power_W`	the power set for it
`T_mean_K`	the average temperature over the heat source: the device temperature
`T_max_K`	the hottest point of the heat source; depends more on the mesh
`dT_mean_K`, `dT_max_K`	the rise above the lowest fixed temperature of the model

In your circuit simulator, set each instance's temperature rise to its `dT_mean_K` (e.g. `dtemp` in ngspice, `trise` in Spectre).

Good to know:

- The temperatures include the heat of all devices. For the self-heating of one device alone, set only its power.
- If the device model computes self-heating too (IHP's npn13G2 model has a thermal resistance, `rth`), the device's own heating counts twice. Then switch the model's self-heating off, or give it only the rise caused by the other devices (a thermal run without this device's power).
- A device's power depends on its temperature. If the temperatures change the powers noticeably, simulate the circuit again with the new temperatures, load the new powers, and run the thermal model once more.

15 Troubleshooting

In this chapter: the messages of the plug-in, what they mean, and what to do.

Message	Meaning	What to do
Stackup: none chosen yet	no stackup for this layout	choose one (chapter 5)
Ground / Devices: needs a stackup	nothing can be checked without a stackup	choose one
no ground: mark a ground shape or add via ports	no via port reaches metal yet	place the pad ports, or mark ground (chapter 7)
short circuit: both ends of via port(s) … on the same net	the port is shorted by the layout	check the port's from/to layers
the marked ground is N separate pieces in the region	the marked ground pieces connect only outside the region	enlarge the region, or check the marks
large region, not checked: Run Checks in the details	checks are skipped for speed	press Run Checks
N cut without ground	the region box cuts signal wires with no ground metal nearby	place those ports by hand
no PDK devices (SG13_dev pcells) in the cell	transistors not recognized	load the IHP PDK in KLayout
N pair(s) of REF patches touch	two devices would share a reference	keep devices apart, or check the layout
N terminal(s) not connected	a device terminal has no wiring in the region	enlarge the region, or check the layout
resistor "not replaced: EM gives R too low"	a resistor stays in the EM model as a sheet of drawn size, without the PDK's width offset and contact end resistance (about 10 % low)	tick include resistors and press Replace All (chapter 9)
port N … not simulated (incomplete)	port settings missing	set them (EM > Port Settings…)
layer '…' is not a metal of the stackup	a port uses a layer the stackup does not know	choose another layer in the port dialog
… is not drawn at the port	the port's from/to layer has no metal at the port	check the port position and layers
All port layers 201..299 are used	no free port layer left	remove unused ports
This layout has not been exported yet	Open in setupEM or EMStudio needs an export first	press Export for EM…
EMStudio ended at once (exit code 1): this EMStudio cannot open a .simcfg	your EMStudio has no .simcfg import yet (it ends on a .simcfg on its command line)	open the export with setupEM, or use an EMStudio with File > Import setupEM Config
no 0 Hz point in the .sNp	the EM result has no DC point	start the sweep at 0 Hz (chapter 11)
Thermal: the stackup … has no layer Activ	the heat sources sit in Activ, which this stackup lacks	choose a stackup for EM and thermal (chapter 14)
Thermal: no devices in the region	the export has no heat sources	choose another region, or draw your own heat sources
Thermal: N without a heat area (bounding box)	a device has no heat shapes and no layers to derive them from	check the device: its whole box heats, the temperature comes out too low
Thermal: N from the PDK's rule	the PDK cells draw no heat shapes (older PDK, static cells)	nothing: the rule gives the PDK's rectangles
Thermal: N cut by the region	a device is partly outside the region: only the part inside heats	enlarge the region
Names: no LVS device: …	LVS has no device at this heat source	run LVS on this cell, or on a cell that contains it once
region outline: choose the target layer (setupThermal)	the placeholder boundary has no height yet	choose its target layer, e.g. BACKSIDEGND (chapter 14)

16 Reference

16.1 Layers used by the plug-in

Layer	Use
201 … 299	ports: port number = layer − 200 (layers used by the stackup itself, e.g. SUBGND, are skipped)
300	REF_FOR_TRANSISTOR: the transistors' and resistors' REF patches
800/0	the EM region box
201 upward (thermal export)	in the exported layout: the region outline, then one heat source per device (stackup layers and your own markers skipped)
401 … 499 (suggested)	your own thermal markers, apart from the EM ports

16.2 Stored in the layout

Saving the layout keeps its EM setup. The export copies none of this into `<name>_em.gds`.

What	Where
Port settings (metals, direction, Z0)	GDS property 200 on each port shape
Boundary port mark	GDS property 201 on ports made by Create Proposed Ports
Replaced device	GDS property 202 on the device's REF patch and ports
Stackup, last export, ground marks	layout meta info `klayout_em_export.stackup`, `.output`, `.ground`
Last thermal export, LVS database	layout meta info `klayout_em_export.thermal_output`, `.lvsdb`

Your original file is kept. The first time the plug-in changes a layout that has none of its data yet (you place a port, choose a stackup, mark ground, replace devices, …), it copies the layout file as it is on disk to `<name>.gds.bak`, next to it. The status bar says so. An existing `.bak` is never replaced. Saving the layout later writes the EM setup into `<name>.gds`; the `.bak` is your original, without any of it. KLayout itself can keep backups of every save as well: **File > Setup > Application > General**, number of backup files.

KLayout saves meta info in an extra cell `$$$CONTEXT_INFO$$$` in the GDS file. KLayout hides it, but other GDS tools may show it as an empty top cell. (KLayout writes this cell anyway for a layout with PCells.)

Before you pass the layout on, e.g. for tape-out, remove the EM setup:

1. Delete the shapes on the port layers, on layer 300 and on 800/0 (see the layer table above). A layout converted with Fill Density Holes and Merge Metal must not be passed on at all: use the original.
2. Remove the meta info in the Python console of the macro editor (**Macros > Macro Development**, F5), then save:

```
import pya
ly = pya.CellView.active().layout()
for key in ("klayout_em_export.stackup", "klayout_em_export.output", "klayout_em_export.ground",
            "klayout_em_export.thermal_output", "klayout_em_export.lvsdb"):
    ly.remove_meta_info(key)
```

The plug-in also keeps a copy of the stackup, the last export and the ground marks in `~/.klayout_em_export.json`. It uses that copy only when the layout has none of its own.

16.3 Files

File	Written by	Content
`<name>_em.gds`	Export	flat layout for the EM simulation
`<name>_em.simcfg`	Export	settings for setupEM and EMStudio
`<name>_em_portmap.json`	Export	ports: layout number, device and terminal, net
`<name>_thermal.gds`	Export for Thermal	flat layout for the thermal simulation
`<name>_thermal.tsimcfg`	Export for Thermal	settings for setupThermal
`<name>_thermal_thermalmap.json`	Export for Thermal	heat sources: layer, device, instance name, heat area
`<model>_temperatures.csv`	setupThermal after Elmer	temperature per heat source and instance
`<model>.sNp`, `<model>_dc.sNp`	gds2palace after Palace	S-parameters, without and with the 0 Hz point
`..._<simulator>.cir` / `.inc` / `.scs`	Create Circuit	the circuit
`..._le.cir` / `_le.inc`	Create Circuit (snp2le)	the EM model's equivalent circuit
`~/.klayout_em_export.json`	the plug-in	remembered choices: stackup, setupEM and EMStudio, Create Circuit settings

16.4 Limits

- Only the IHP SG13G2 PDK cells are recognized as devices (npn13G2 family, nmos/pmos and their HV and RF variants, rsil, rppd, rhigh).
- The 1 mH ties are ideal: the wiring's DC resistance is not in the bias (in the cascode it would lower Icc by about 6 %).
- Resistors from static cells (no PDK parameters) become ideal resistors of their label value: no temperature behaviour, no parasitics. A static npn cell (e.g. `npn13G2$1` in a layout saved without pcell context, as some fill tools write it) gets its size from its `Ae=Nx*Ny*we*le` label; without that label it is not replaced.
- Resistors left in the EM model (not replaced) come out too low: the sheet has the drawn size, without the PDK's width offset and contact end resistance. rhigh and rppd are about 10 % to 13 % low, rsil up to about 2 % (chapter 9). A resistor of several segments cannot be replaced yet.
- The REF patch adds a few fF per transistor when the device is wired directly (chapter 13).
- Thermal: instance names from LVS are tested with npn transistors and resistors. MOSFETs are supported, but not tested yet.
- Thermal: for the cascode's Q1 (npn13G2, 8 emitters), the thermal model with a converged mesh gives about 25 % more temperature rise than the `rth` of IHP's compact model. The reason is not known yet. Use the temperatures as estimates, and for comparing devices and layouts.